



Statistical learning and exchange rate forecasting

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ABSTRACT

This study uses innovative tools recently proposed in the statistical learning literature to assess the capability of standard exchange rate models to predict the exchange rate in the short and long runs. Our results show that statistical learning methods deliver remarkably good performance, outperforming the random walk in forecasting the exchange rate at different forecasting horizons, with the exception of the very short term (a period of one to two months). These results were robust across countries, time, and models. We then used these tools to compare the predictive capabilities of different exchange rate models and model specifications, and found that sticky price versions of the monetary model with an error correction specification delivered the best performance. We also explain the operation of the statistical learning models by developing measures of variable importance and analyzing the kind of relationship that links each variable with the outcome. This gives us a better understanding of the relationship between the exchange rate and economic fundamentals, which appears complex and characterized by strong non-linearities.

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1. Introduction

A well-known puzzle in international finance is the inability of standard models to forecast the exchange rate, both in the short and long runs. This is linked to another major puzzle known as the “exchange rate disconnect puzzle”. Although not equivalent, the two puzzles are strongly related. The exchange rate forecasting puzzle dates back to the seminal work by Meese and Rogoff (1983), who showed that standard models of exchange rate are systematically outperformed by a random walk in out-of-sample forecasting. The second puzzle refers to “the exceedingly weak relationship (except, perhaps, in the longer run) between the exchange rate and virtually any macroeconomic aggregates” (Obstfeld & Rogoff,

2001), despite the presence of a vast amount of theoretical literature that posits a strong relationship between the exchange rate and macroeconomic fundamentals.

One possible explanation for the empirical failure of standard models in predicting the exchange rate is that a relationship does obtain between exchange rates and fundamentals, but such a relationship is complex and poorly captured by currently available tools. These are generally linear models that allow for limited interactions among the variable and pose too many constraints, resulting in poor forecasting performance.

This study addresses these problems using tools that have recently appeared in the statistical learning literature. Our aim is twofold. On the one hand, we assess whether statistical learning (sl) methods applied to standard exchange rate models are useful in (out-of-sample) forecasting of the exchange rate.² This can provide

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² Most people, among practitioners but also in the scientific community, use the terms statistical learning and machine learning interchangeably. Given the nature of our exercise (we are not only

evidence for the *existence* of a relationship between exchange rates and macroeconomic fundamentals. On the other hand, by investigating the “black box” of *SL* tools, we aim to improve our understanding of the *types* of links between macroeconomic fundamentals and the exchange rate implicit in statistical learning models.

We innovate with respect to the literature in several domains. First, we apply the most promising and innovative tools in the *SL* literature to a set of rather standard exchange rate models. Our objective is not limited to the out-of-sample prediction of the exchange rate *per se*, but rather extends to the assessment of exchange rate models through recent *SL* methods. Second, we show that, with the exception of the very short run (one to two months), *SL* models display remarkably good predictive abilities, consistently outperforming the random walk benchmark. These results were robust across countries, time, models, and forecasting horizons. Based on them, we can confirm that the fundamentals do matter for exchange rate forecasting. Third, among our selection of *SL* algorithms, support vector machines delivered the best performance. Fourth, we do not limit ourselves to a horse-race between statistical learning methods. Instead, we use these methods to compare the predictive capabilities of different exchange rate models and model specifications. We focus on regressors implied by four exchange rate models to predict the forward difference in exchange rates using the explanatory variables in difference or in levels. In the latter case, we assume that some kind of non-linear error correction mechanism (ECM) may hold. Overall, we found that the sticky price version of the monetary model in ECM form achieves the best performance. Finally, we take a further step in explaining the behavior of statistical learning tools. We examine them by developing measures of variable importance consistent with our prediction scheme, and by representing the partial relationship linking each regressor with the response variable. This is a major improvement over current economic literature, where *SL* methods are generally treated as black boxes incapable of producing information on factors that contribute the most to predictive performance.

The remainder of the paper is structured as follows: Section 2 presents a brief overview of the relevant literature, Section 3 shows the motivation for the need to use statistical learning tools in the context of exchange rate forecasting, Section 4 presents the methodology and data, Section 5 illustrates the results, and Section 6 offers the conclusions of this study.

2. Related literature

The early literature on exchange rate modeling dates back to work by Meade (1951) with the development of the Keynesian framework. However, it was with contributions by Mundell (1962, 1963) and Fleming (1962) that the Keynesian open economy model became the dominant framework in the literature. The demise of the

Bretton Woods system spurred a new debate on both the theoretical and empirical sides of the puzzle. On the theoretical side, the original Mundell–Fleming model has been modified to incorporate expectations in a truly asset market-based approach in a flexible (Frenkel, 1976; Mussa, 1976) or sticky price (Dornbusch, 1976; Wilson, 1979) setting. The standard model has also been modified to allow for imperfect substitutability between domestic and foreign assets (Branson & Henderson, 1985). However, as soon as a sufficient amount of data from the floating period started to become available, exchange rate models struggled to find consistent empirical support, especially in out-of-sample forecasting. In particular, the empirical results were invalidated by the seminal contribution by Meese and Rogoff (1983), who found that standard exchange rate models had no predictive ability. Since then, outperforming the random walk has been a formidable challenge for any exchange rate model in most common forecasting intervals. On the theoretical side, advances have been made over time, with new Keynesian models (Obstfeld & Rogoff, 2001) providing microfoundations for open economy models; however, on the empirical side, the challenges remained unanswered. Some models, such as the Taylor rule (Molodtsova & Papell, 2009), appear to be more effective in the short run, and monetary models display some predictive power in the long run (Cheung, Chinn, & Pascual, 2005; Mark, 1995), but the general picture is rather grim. In the words of Rossi (2013), “none of the predictors, models, or tests systematically finds empirical support of superior exchange rate forecasting ability across all countries and time periods: When predictability appears, typically it does so occasionally for some countries and for short periods of time”.

In asset prices models it may be natural to observe such empirical failures. Engel and West (2005) argued that if the discount factor is close to one and the fundamentals follow a near unit root process, the exchange rate should manifest a near-random walk behavior resulting in poor predictability, especially at short horizons.³

Nonetheless, most economists would agree that the relationship between the exchange rate and the fundamentals postulated by economic models should translate into models that can predict the exchange rate to some degree. This explains the almost obsessive interest of the economic literature in this topic: A simple Google Scholar search for “exchange rate forecast” yields more than 154,000 hits.⁴

3. Why statistical learning models?

One possible explanation for the empirical failure of standard exchange rate models to predict the exchange rate is that the tools used are not powerful and flexible enough to capture the complex relationship between the exchange rate and fundamentals. For example, a growing

interested in pure prediction, but also in model comparison and the relationship between variables), throughout this paper, we use the term statistical learning.

³ In the long run, some studies have found evidence of cointegration between the exchange rate and fundamentals (MacDonald & Marsh, 1997).

⁴ The search was conducted on September 5, 2018.

body of literature has highlighted the presence of non-linearities in exchange rate dynamics (Sarno, Valente, & Leon, 2006; Taylor, Peel, & Sarno, 2001).

Statistical learning tools are extremely flexible methods that detect patterns in data and use them to make predictions. These patterns may be highly non-linear and based on the interaction among variables.⁵ In the jargon of the SL literature, we use supervised learning techniques with the goal of predicting the value of a response variable (future increments in the exchange rate) using a set of input variables or *features* (fundamentals). “Supervised” refers to the fact that the presence of an observable response variable guides the learning process.⁶ SL tools are well known in the scientific community and have been applied to several fields, such as engineering, physics, and biology, since the early 90 s. More recently, these tools have also gained popularity in the social sciences following the availability of large datasets on which they can be trained.⁷ Most applications of SL to economics have been for purely predictive purposes,⁸ as these tools are extremely efficient in prediction.⁹ Therefore, given the inability of standard models to predict the exchange rate, this is a perfect setting in which to apply SL methods.

However, the purpose of this study is not only prediction. We also use SL to improve our understanding of the exchange rate and its relationship with economic fundamentals. In particular, we use SL methods to compare the predictive power of different exchange rate models and different model specifications. We then open the black boxes of these methods, unveiling interesting findings, such as variable importance and the form of the interaction between the fundamentals and changes in the exchange rates.¹⁰

Confining our study to standard exchange rate models also allows us to eliminate possible misconceptions about the application of SL to financial markets. As these tools are extremely efficient at detecting patterns in data,

in principle, they can capture the effect of any non-fundamental determinant of the exchange rate, the effect of possible non-rational behavior by market participants, and that of such external factors as central bank intervention. In this context, statistical learning may be considered a sophisticated variant of technical analysis. To preclude this type of interpretation, we apply SL to only traditional exchange rate models, thus excluding past (lagged) prices and exchange rates from which further patterns can be detected. Therefore, in our setting, SL algorithms can be viewed merely as flexible non-parametric statistical methods applied to otherwise standard exchange rate models.

4. Methods and data

4.1. Exchange rate models

Before we describe the models, we introduce some notation: Let Δ_k be the k -period (backward) difference (e.g., $\Delta_k x_t = x_t - x_{t-k}$), e_t denote the (log of) the exchange rate expressed in US dollars at the end of period t , and X_t be a set of fundamentals (to be defined below). Finally, let a “hat” over a variable denote cross-country differentials (the United States was the reference country in our case); for example, $\hat{y}^i = y^i - y^{US}$.

We use different models of the exchange rate with different sets of variables as fundamentals. For each model, following Cheung et al. (2005) and Cheung, Chinn, Pascual, and Zhang (2018), we estimate two versions: one is a model in the difference form and the other an error correction model. The latter approach allows the model to better incorporate the notion of the equilibrium exchange rate and the relative adjustment process. In both cases, we compute predictions at various intervals, starting from the very short run (one month ahead) to the long run (12 months ahead).

The difference model is the nonlinear version of

$$\Delta_k e_{t+k} = \alpha + \Delta_k X_t' \gamma + \epsilon_{t+k}, \quad (1)$$

that is,

$$\Delta_k e_{t+k} = f(\Delta_k X_t) + \epsilon_{t+k}, \quad (2)$$

where $f()$ is the unknown function to estimate. The error correction model is the non-linear version of

$$\Delta_k e_{t+k} = \alpha + \beta(e_t - X_t' \tilde{\gamma}) + \epsilon_{t+k}, \quad (3)$$

that is,

$$\Delta_k e_{t+k} = f(e_t, X_t) + \epsilon_{t+k}, \quad (4)$$

where, again, $f()$ is an unknown function to estimate. In the non-linear model, one has to estimate the entire function $f()$, and not only the regression coefficients.

In the equations above, the term X contains variables that identify different models. In particular, using the symbols m , y , π , and i , respectively, for the log of money, log of output, log of price level, and interest rate, we consider the following models:

1. Standard monetary model: $X_t = [\hat{m}_t, \hat{y}_t]$
2. Monetary model with sticky prices: $X_t = [\hat{m}_t, \hat{y}_t, \hat{\pi}_t]$

⁵ Coulombe, Leroux, Stevanovic, and Surprenat (2019) showed that allowing for non-linearity and interactions is one of the major advantages of statistical learning tools for macroeconomic forecasting.

⁶ In unsupervised learning, there is no observable response variable, and the techniques are designed to detect possible patterns that emerge in the input variables.

⁷ See Kotchoni, Leroux, and Stevanovic (2019) for a discussion of statistical learning models in a data-rich environment.

⁸ There is an emerging literature on SL and casual inference has been surveyed well by Athey (2018).

⁹ See Colombo, Forte, and Rossignoli (2018), Huerta, Corbacho, and Elkan (2013), and Papadimitriou, Gogas, and Stathakis (2014), Gonz  les Rojas and Herman (2019) for financial market forecasts, and Holopainen and Sarlin (2017) for an application to early warning indicators.

¹⁰ Our study differs substantially from that by Amat, Michalski, and Stoltz (2018), who tested the predictive power of SL tools for exchange rate forecasting, and from that by Li, Tsiakas, and Wang (2015), who applied an elastic net to a kitchen-sink regression of all possible lagged fundamental variables. With respect to these studies, we applied a larger set of SL models to allow for non-linearity and interactions in well-established exchange rate models. But, most importantly, we are not interested in simply finding the best predictive model using whatever information is available. Our objective is to explain SL models, and use them to improve our understanding of the empirical relationship between the exchange rate and fundamentals.

3. Monetary model with sticky prices and UIP deviations: $X_t = [\hat{m}_t, \hat{y}_t, \hat{\pi}_t, \hat{i}_t]$
4. Taylor rule model: $X_t = [\tilde{y}_t, \hat{\pi}_t, \hat{i}_t]$ where $\tilde{y}$ defines the output gap.

In the next section, we follow most of the literature on estimating rolling regressions in advanced economies in the post-Bretton Woods period. The estimation (training) sample was set to 120 observations (10 years), after which out-of-sample forecasts were produced. Then, the sample was rolled forward one period at a time, and the procedure was repeated until no more observations were available to forecast. This approach is generally preferred over recursive forecasts¹¹ as it allows for the consideration of parameter instability over time.

4.2. Methods

In addition to the standard regression approach, we use *sl* algorithms, focusing in particular on the most widely used techniques. In the following paragraphs, we provide a brief introduction to these techniques. For details, see the excellent monographs by [Hastie, Tibshirani, and Friedman \(2009\)](#) and [James, Witten, Hastie, and Tibshirani \(2013\)](#).

In implementing alternative approaches, we choose models that share the following features:

1. Flexibility: Exchange rate forecasts may require regression models or classification models depending on whether one is interested in forecasting the magnitude or the direction of change of the exchange rate.
2. Non-linearity: The function mapping the explanatory variables to the response variable may be non-linear.
3. Interactions: Standard exchange rate models generally assume reduced forms, where variables are entered additively, whereas more complex models may consider possible interaction effects. For example, the effect of money or interest rate on the exchange rate may depend on the level (or the growth rate) of GDP.

In this study, we use the major *sl* tools that share the above properties: regularized regression splines with and without interactions, random forests, and support vector machines.

4.2.1. Regularized regression splines

Throughout the next section, we consider a generic relationship between the exchange rate e or its increment (dependent variable), and fundamentals x (regressors/predictors). We start by assuming that x is scalar, and observe a sample of data $\{(e_1, x_1), \dots, (e_n, x_n)\}$ generated by the process

$$e_t = f(x_t) + \epsilon_t, \quad t = 1, 2, \dots, n,$$

where ϵ_t are independent random variables with zero mean and finite variance σ^2 , x_t are either given numbers or realizations of a random variable independent of ϵ_t , and $f(\cdot)$ is an unknown function. Estimating $f(\cdot)$ using cubic splines entails the following two steps.¹²

Step 1 fixing K points (*knots*) so that the x_t are partitioned into $K + 1$ contiguous intervals; and

Step 2 at every interval, fitting the least-squares regression

$$e_t = \beta_{0j} + \beta_{1j}x_t + \beta_{2j}x_t^2 + \beta_{3j}x_t^3 + \epsilon_t, \quad j = \{0, 1, 2, \dots, K+1\}$$

with the constraint that on every knot ξ_k , the curve $g(x_t) = \beta_{0j} + \beta_{1j}x_t + \beta_{2j}x_t^2 + \beta_{3j}x_t^3$ must be continuous with its first two derivatives.

Because the space of cubic spline functions on a given sequence of knots is a vector space, there are many equivalent bases for representing them. The *B-spline* is a computationally convenient basis for splines. In synthesis, we want to approximate the unknown function $f(\cdot)$ using a spline function $g(\cdot)$ that can be represented as linear combination of known basis functions

$$g(x) = \sum_{j=1}^{K+4} \phi_j g_j(x),$$

with ϕ_j (projection) coefficients. The set of B-spline functions $g_j(x)$ for x taking values in $[0, 1]$ and knots $\{0.2, 0.4, 0.6, 0.8\}$ are depicted in [Fig. 1](#). From a practical point of view, Step 2. consists of simply running a regression of the response e_i on the $K + 4$ new variables $g_j(x)$.

The approach can be generalized to the case of multiple regressors by applying the B-splines to each variable and, possibly, including all interactions between the basis functions. For example, if we consider the two regressors $\hat{m}_t$ (log-money) and $\hat{y}_t$ (log-output), we can approximate the unknown function $f(\hat{m}_t, \hat{y}_t)$ using the additive model¹³

$$e_t = \sum_j \psi_{0j} g_j(\hat{m}_t) + \sum_k \psi_{k0} h_k(\hat{y}_t) + \epsilon_t, \quad (5)$$

with $g_j(\cdot)$ and $h_k(\cdot)$ B-splines, or by using the model with all interactions

$$e_t = \sum_j \psi_{0j} g_j(\hat{m}_t) + \sum_k \psi_{k0} h_k(\hat{y}_t) + \sum_j \sum_{k < j} \psi_{jk} g_j(\hat{m}_t) h_k(\hat{y}_t) + \epsilon_t. \quad (6)$$

This approach can be easily extended to an arbitrary number of regressors at the cost of an increasing number of coefficients to be estimated. We refer to the above two models as *cubic spline regression* (5) and *cubic spline regression with interactions* (6), respectively.

Given the complexity (i.e., the large number of degrees of freedom of the model) of these models and their tendency to produce high-variance predictions, we regularize the estimation by elastic net penalization.

¹¹ Here, starting from the test period, the sample is simply extended one period at a time instead of being rolled.

¹² See [Hastie et al. \(2009\)](#) Ch.5 for a detailed explanation.

¹³ See the section on additive models in [Hastie et al. \(2009\)](#).

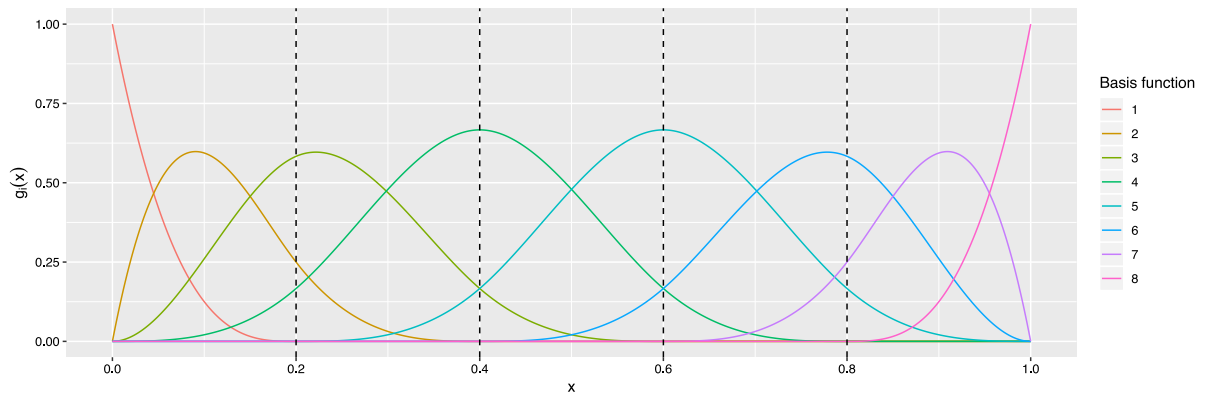


Fig. 1. Cubic B-splines basis for the knots: 0.2, 0.4, 0.6, 0.8.

When using highly flexible non-parametric models, one needs to consider the bias–variance trade-off problem. Consider a general statistical model $e = f(x) + \epsilon$, for which we need to assess the out-of-sample predictive accuracy using the mean squared error (MSE) as a risk measure. Given an out-of-sample observation x_0 and an estimation sample (which does not include x_0) used to obtain the fitted model $\hat{f}$, it can be shown that MSE can be decomposed as follows:

$$E(y_0 - \hat{f}(x_0))^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon). \quad (7)$$

The term $\text{Var}(\hat{f}(x_0))$ refers to the variance $\hat{f}(x_0)$ as an estimator of $f(x_0)$, in a repeated sampling framework; $\text{Bias}(\hat{f}(x_0))$ denotes the bias of the estimator, that is, the difference between the average prediction of the model and the true value. Eq. (7) shows that minimizing the prediction error requires selecting a model with both a low variance and a low bias. Statistical learning models, as they are very flexible and allow for non-linearities, can achieve a low bias but at the cost of a high variance. One solution is to introduce some sort of *regularization* that reduces variance at the cost of introducing a small bias. Ridge and Lasso are well-known forms of regularization that work by adding a penalty term proportional to some norm of the vector of coefficients to the sum of squares (for least-squares estimation) or negative log-likelihood (for maximum likelihood estimation). Ridge regression regularizes the coefficients by adding an L_2 penalty, whereas Lasso uses an L_1 penalty, which tends to produce a sparse solution by setting some coefficients to zero. In a highly influential paper, [Zou and Hastie \(2005\)](#) introduced the *elastic net* penalty, and showed that it can improve both Lasso and Ridge regression.

$$\lambda \sum_{j=1}^p (\alpha \beta_j^2 + (1 - \alpha) |\beta_j|) \quad (8)$$

where p is the number of estimated coefficients. As $\alpha \rightarrow 1$, the penalty collapses to the Ridge penalty whereas as $\alpha \rightarrow 0$, it collapses to the Lasso penalty.

In our setting, where the response variable is the increment in the log-exchange rate $\Delta_k e_{t+k}$, the regularized estimators have a natural interpretation. Indeed, by

shrinking the regression coefficients toward zero, the estimated model is biased in the direction of the random walk, which is the standard benchmark model in exchange rate forecasting.

In our application the tuning parameters were selected using 10-fold cross-validation applied to the first estimate of the 10-year rolling window.¹⁴ The parameters selected on the initial window were used on all subsequent windows. Thus, the tuning parameters were possibly different for every model and country.

4.2.2. Random forest

Random forests are derived from regression trees, and constitute their evolution and improvement. Regression trees are known in statistical learning for handling complex problems involving interactions and non-linearities. Moreover, the same tool can be used to handle both quantitative responses (regression trees) and qualitative ones (classification trees). Regression trees are constructed by dividing the predictor space into a set of J non-overlapping regions R_j . Subsequently, in each region, the mean response values for the training observations are computed. The selection of the regions (internal nodes) $1, \dots, J$ is determined by minimizing the residual sum of squares (RSS). This is done recursively by considering all possible predictors $X_1, \dots, X_p$ and all possible cutoff points s , and selecting the predictor–cutoff point pair that delivers the greatest reduction in the RSS, in other words, we choose the predictor j and cutoff point s that minimize

$$\sum_{i: x_i \in R_1(j, s)} (e_i - \hat{e}_{R_1})^2 + \sum_{i: x_i \in R_2(j, s)} (e_i - \hat{e}_{R_2})^2 \quad (9)$$

where R_1 and R_2 are the regions of the space of predictor j , and $\hat{e}_{R_1}$ is the mean response of the training observations in the $R_1(j, s)$ region of predictor j , defined by the cutoff point s ; $\hat{e}_{R_2}$ is defined analogously. Subsequently, the process is repeated recursively for each of the remaining predictors.

In complex problems, decision trees are known to grow large, and suffer from instability (i.e., adding or replacing few observations can result in large modifications)

¹⁴ We used the function `cv.glmnet()` of the R package *glmnet* by [Friedman, Hastie, and Tibshirani \(2010\)](#).

Table 1
1-month forecast. Monetary model.

	First difference specification					ECM specification				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	1.05	1.00	1.00	1.17	1.10	1.06	1.00	1.00	1.18	1.07
AT	1.09	0.98	0.98	1.33	1.13	1.08	0.99	0.99*	1.07	1.01
BE	1.06	1.00	1.00	1.24	1.14	1.17	1.00	1.01	1.11	1.15
CA	1.04	1.01	1.01	1.17	1.05	1.06	1.01	1.01	1.21	1.12
DK	1.05	1.00	1.00	1.25	1.04	1.05	1.00	1.26	1.15	1.10
FI	1.05	1.00	1.00	1.28	1.08	1.03	1.00	1.00	1.14	1.09
FR	1.09	1.00	1.00	1.25	1.09	1.07	1.00	1.00	1.15	1.07
DE	1.07	1.01	1.00	1.33	1.17	1.20	1.03	1.05	1.18	1.04
GR	0.97	1.00	1.00	1.29	1.03	1.01	0.99	0.98*	1.30	0.97
IE	1.08	1.01	1.01	1.17	1.17	1.18	1.01	1.01	1.24	1.09
IT	0.98	1.00	1.00	1.20	1.02	1.10	1.00	1.00	1.20	1.04
JP	1.02	1.00	1.00	1.22	1.10	1.06	1.00	1.00	1.18	1.09
NL	1.05	0.99	0.99	0.99	1.08	1.03	0.99	0.99*	1.36	1.06
NZ	1.01	1.00	1.00	1.11	1.01	1.05	1.00	1.00	1.18	0.95
PT	0.98	1.00	1.00	0.99	0.98	1.05	1.00	0.99	1.20	1.09
ES	1.05	1.01	1.01	1.21	1.09	1.10	1.02	1.02	1.28	1.11
SE	1.07	1.00	1.00	1.29	1.10	1.11	1.01	1.01	1.15	1.06
CH	1.05	1.00	1.00	1.13	0.99	1.05	1.00	1.00	1.09	1.17
GB	1.04	1.00	1.00	1.40	1.07	1.09	1.00	1.00	1.14	1.12

The table reports the forecasting abilities of different methods: standard ordinary least squares (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the ratio of the root mean-squared forecast error of the model relative to that of the random walk without drift: Values smaller than unity denote that the model's forecasts were better than the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one with the lowest RMSE ratio (when significant). The forecasts were for one month using the monetary model.

and high variance. It is also well known that in the presence of independent sets of observations, each with its own variance, averaging across all the sets of observations reduces variance. Random forests constitute an evolution of regression trees that allows to considerably reduce the variance in the predictions. This is done by bootstrapping the training samples to build a number of decision trees. However, when building these trees, only a random subset of potential predictors is proposed at each new split. This allows for the *decorrelation* of the trees, substantially reducing variance.¹⁵ For a detailed description of random forests, the interested reader can refer to [Hastie et al. \(2009, Ch.15\)](#).

Random forests have few tuning parameters that are either fixed according to some rule or determined through cross-validation (CV). The only tuning parameters we determined by using CV were the number of variables randomly sampled as candidates at each split and the maximum number of nodes in each tree. We tried different values on the first 10-year estimation window of different countries, and picked two for the former and five for the latter. All other tuning parameters were set to default values.¹⁶

¹⁵ The simple bootstrap procedure tends to produce highly correlated trees, which, when averaged out, do not contribute to a large reduction in variance.

¹⁶ We used the `randomForest()` function of the R package *randomForest* ([Liaw & Wiener, 2002](#)), which is based on Breiman and Cutler's original FORTRAN code.

Table 2
1-month forecast. Monetary model sticky prices.

	First difference specification					ECM specification				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	1.06	1.00	1.00	1.11	1.14	1.08	1.00	1.00	1.19	1.06
AT	1.14	0.98	0.98	1.19	1.14	1.10	0.99*	0.99*	1.01	1.00
BE	1.08	1.01	1.01	1.17	1.13	1.23	1.00	1.35	1.14	1.15
CA	1.04	1.01	1.01	1.14	1.10	1.06	1.01	1.01	1.17	1.08
DK	1.07	1.00	1.00	1.13	1.14	1.03	1.00	1.26	1.08	1.08
FI	1.05	1.00	1.00	1.15	1.16	1.06	1.00	1.00	1.13	1.07
FR	1.10	1.00	1.00	1.15	1.13	1.08	1.00	1.00	1.20	1.06
DE	1.08	1.01	1.01	1.15	1.15	1.29	1.01	1.08	1.17	1.03
GR	0.97	1.00	1.00	1.18	1.01	1.03	0.97*	1.01	1.32	0.98
IE	1.11	1.00	1.00	1.18	1.16	1.16	1.01	1.15	1.18	1.08
IT	1.00	1.00	1.00	1.14	1.08	1.03	1.10	1.06	1.15	1.07
JP	1.03	1.00	1.00	1.14	1.11	1.09	1.00	1.00	1.14	1.10
NL	1.09	0.98	0.98	1.06	1.08	1.03	0.99*	0.99*	1.33	1.00
NZ	1.02	1.00	1.00	1.01	1.06	1.07	1.00	1.00	1.16	0.94
PT	1.03	1.00	1.00	1.01	1.03	1.08	0.99	0.99	1.21	1.10
ES	1.06	1.01	1.01	1.15	1.11	1.03	1.02	1.02	1.24	1.08
SE	1.10	1.00	1.00	1.23	1.13	1.15	1.01	1.01	1.14	1.07
CH	1.07	1.00	1.00	0.97	1.15	1.05	1.10	1.00	1.13	1.24
GB	1.06	1.00	1.00	1.24	1.08	1.12	1.00	1.00	1.13	1.11

The table reports the forecasting abilities of different methods: standard ordinary least squares (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the ratio of the root mean-squared forecast error of the model relative to that of the random walk without drift: Values smaller than unity denote that the model forecasts were better than the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one with the lowest RMSE ratio (when significant). The forecasts were for one month using the monetary model with sticky prices.

4.2.3. Support vector machine

Support vector machines (svms) are one of the most popular *SL* algorithms due to their efficiency of predictive regression and classification.

An svm is an algorithm that constructs a hyperplane that maximizes the distance between classes of observations (for example positive and negative exchange rate changes). When the two classes are linearly separable, an svm is analogous to a linear optimization problem solvable with standard tools (i.e., Lagrange multipliers). Often, however, linear separation is not possible; in these cases, a useful result from [Cortes and Vapnik \(1995\)](#) shows that it is possible to project the dataset through a kernel function onto a higher-dimensional space (i.e., the feature space), where the dataset is linearly separable.

An svm selects the optimal separating hyperplane in the following way: First, the perpendicular distance between any observation and the given hyperplane is computed. Second, the smallest distance between the observation and the hyperplane is identified, which defines the margin, that is, the maximal width of the slab parallel to the hyperplane that has no interior data points. The optimal separating hyperplane maximizes the margin.

More formally, let $\mathbf{x}$ be a p -element vector of variables and let n be the number of training observations. The svm is the classification function ([James et al., 2013](#))

$$f(\mathbf{x}) = \sum_{i=1}^n \alpha_i K(\mathbf{x}, \mathbf{x}_i) + \beta_0 \quad (10)$$

Table 3

1-month forecast. Monetary model sticky prices, UIP deviations.

	First difference specification					ECM specification				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	1.07	1.01	1.01	1.13	1.14	1.09	1.00	1.00	1.16	1.05
AT	1.14	0.98	0.98	1.25	1.18	1.12	0.99	0.98	1.06	1.00
BE	1.08	1.01	1.01	1.17	1.10	1.26	1.00	1.25	1.25	1.09
CA	1.04	1.01	1.01	1.11	1.10	1.06	1.01	1.01	1.14	1.10
DK	1.03	1.00	1.00	1.28	1.22	1.04	1.00	1.74	1.13	1.17
FI	1.10	1.01	1.01	1.22	1.26	1.02	1.00	1.00	1.09	1.16
FR	1.20	1.00	1.00	1.11	1.11	1.10	1.00	1.00	1.26	1.03
DE	1.08	1.01	1.01	1.10	1.13	1.31	1.01	1.08	1.13	1.06
IE	1.24	1.00	1.02	1.20	1.16	1.20	1.01	1.01	1.15	1.06
IT	1.02	1.00	1.00	1.06	1.14	0.97	1.00	1.00	1.13	1.11
JP	1.03	1.00	1.00	1.08	1.13	1.07	1.00	1.00	1.17	1.10
NL	1.18	0.98	0.98	1.14	1.14	1.10	0.99	0.99**	1.20	1.01
NZ	1.02	1.00	1.00	1.04	1.07	1.10	1.00	1.00	1.13	0.97
PT	1.04	1.00	1.00	0.96	1.03	0.99	1.00	1.01	1.22	1.11
ES	1.07	1.01	1.01	1.12	1.15	1.04	1.02	1.02	1.25	1.08
SE	1.24	1.00	1.00	1.20	1.13	1.14	1.01	1.01	1.12	1.07
CH	1.08	1.00	1.00	0.90	1.10	1.08	1.01	1.00	1.11	1.28
GB	1.07	1.00	1.00	1.18	1.08	1.15	1.00	1.00	1.13	1.06

The table reports the forecasting abilities of different methods: standard ordinary least squares (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), and SVMs (SVM). For each method is displayed the ratio of the root mean-squared forecast error of the model to that of random walk without drift: Values smaller than unity denote that the model forecasted better than the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., that exhibited the lowest RMSE ratio (when significant). The forecasts were for one month, and used the monetary model with sticky prices and UIP deviations.

where $\mathbf{x}_i$ is the vector of variables in the training set, β_0 is a constant that shifts the output of the svm, and α_i are parameters that are non-zero only for the support vectors, and depend on the tuning parameter for soft margin classification. Both β_0 and α_i are chosen optimally within the training period. Finally, we choose the kernel function K following the literature, selecting the radial kernel:

$$K(x, x_i) = \exp(-\gamma \|x - x_i\|^2) \quad (11)$$

where γ is a parameter chosen to optimize the fit of the model to the sample.

In practice, when a test observation x is far from a training observation x_i , this implies that over the p dimensions, the Euclidean distance $\|x - x_i\|$ is large and $K(x, x_i)$ is extremely small. Therefore, the training observation x_i plays no role in predicting the class for the test observation x . Using the kernel function is computationally effective because the kernel has to be computed for each possible pair x, x_i , that is, $n(n-1)/2$ pairs.

svm's have also been adapted to regression problems with a loss function

$$\ell_\epsilon(r) = \begin{cases} 0 & |r| \leq \epsilon \\ |r| - \epsilon & |r| > \epsilon. \end{cases}$$

svm's have several interesting features that make them useful for exchange rate forecasting. The two main advantages are the following:

1. By construction, an svm is optimized to discriminate around the decision margin while attaching no

Table 4

1-month forecast. Taylor rule model.

	ECM specification				
	REG	SP	SPI	RF	SVM
AU	1.07	1.00	1.00	1.09	1.04
AT	1.16	1.00	1.00	1.07	1.07
BE	1.08	1.01	1.00	1.07	1.06
CA	1.15	1.01	1.01	1.16	1.16
DK	1.15	1.00	1.00	1.08	1.06
FI	1.16	1.00	1.00	1.08	1.13
FR	1.09	1.01	1.01	1.11	1.07
DE	1.18	1.01	1.00	1.10	1.11
IE	1.18	1.00	1.00	1.06	1.08
IT	1.06	1.00	1.00	1.05	1.03
JP	1.09	1.00	1.00	1.06	1.11
NL	1.12	1.01	1.01	1.06	1.08
NZ	1.12	1.00	1.00	1.15	1.08
PT	1.12	1.00	1.00	1.09	1.11
ES	1.05	1.00	1.00	1.12	1.07
SE	1.23	1.00	1.00	1.09	1.10
CH	1.83	1.00	1.00	1.11	1.48
GB	1.12	1.00	1.00	1.10	1.14

The table reports the forecasting abilities of different methods: standard ordinary least squares (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the ratio of the root mean-squared forecast error of the model to that of the random walk without drift: Values smaller than unity denote that the model forecasted better than the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one with the lowest RMSE ratio (when significant). The forecasts were for one month and used the Taylor rule model.

weight to easily classifiable data. This is the main difference between it and a regression-based approach: The latter weighs all observations, and not just those close to the decision margin, thus making the regression-based approach less effective on classification problems.

2. svm's deal with non-linearity naturally through the kernel function without imposing a particular functional form, which is valuable in cases such as ours as there is no well-established theory providing clearly testable implications. This can help solve a general problem with non-linear models, which are known to perform well in in-sample but fail in out-of-sample forecasting (Teräsvirta, 2006).

In our application, the svm required selecting two major elements. The first was the kernel. As stated above (Eq. (11)), we selected the radial kernel, following the literature. The second element was the parameter γ that controls the trade-off between bias and variance in the model. By using cross-validation implemented on the first 10 years of the data available, we set $\gamma = 0.5$ for one-month forecasts and $\gamma = 2$ for 12-month forecasts.¹⁷

4.3. Assessing model performance

To assess model performance, we follow the conventional approach in the literature, which focuses on

¹⁷ We used the `svm()` function of the R package *e1071* (Chang & Lin, 2019; Meyer, Dimitriadou, Hornik, Weingessel, & Leisch, 2018).

Table 5
12-month forecast. Monetary model.

	First difference specification					ECM specification				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	0.92	0.88**	0.83***	0.80***	0.78***	0.79***	0.78***	0.68***	0.62***	0.35***
AT	0.71***	1.68	0.86	0.78	0.46***	0.60***	0.74*	0.35***	0.54***	0.20***
BE	1.44	1.07	1.28	1.35	0.76**	1.06	0.54***	0.30***	0.50***	0.20***
CA	1.08	1.03	1.04	0.90**	0.79***	0.92*	0.98	0.91***	0.59***	0.34***
DK	0.98	0.95	0.91**	0.84***	0.62***	0.84*	0.54***	0.28***	0.56***	0.24***
FI	1.10	1.02	1.01	0.99	0.96	0.75***	0.71***	0.54***	0.44***	0.33***
FR	1.01	0.94	0.81***	0.68***	0.40***	0.65***	0.59***	0.36***	0.47***	0.32***
DE	1.51	1.13	1.09	1.17	0.77***	1.30	0.75**	0.36***	0.49***	0.16***
GR	0.63***	0.59***	0.61***	0.68**	0.48***	0.60***	0.72***	0.72***	0.54***	0.29***
IE	1.62	1.31	1.34	1.48	1.63	1.52	1.26	0.53***	0.70***	0.30***
IT	0.96	0.96***	0.89***	0.98	0.85*	1.09	0.49***	0.55**	0.55***	0.32***
JP	0.88***	0.91***	0.72***	0.78***	0.69***	0.92*	0.61***	0.36***	0.43***	0.23***
NL	0.97	0.50***	1.30	0.85	0.42***	0.53***	1.02	0.66**	0.51***	0.23***
NZ	0.93	0.79***	0.72***	0.81**	0.74**	0.92	0.38***	0.47***	0.49***	0.27***
PT	0.90	0.92***	0.82	0.64**	0.75	0.79**	0.70***	0.72***	0.60***	0.37***
ES	1.51	1.30	1.28	1.24	1.07	1.25	1.08	1.10	1.03	0.56***
SE	1.14	1.07	1.08	0.99	1.00	1.12	0.89**	0.83***	0.65***	0.28***
CH	0.40***	0.77***	1.02	0.60***	0.30***	0.86	1.07	0.37***	0.61**	0.25***
GB	1.04	0.91**	0.84***	0.83***	0.84***	1.00	0.80***	0.59***	0.57***	0.43***

The table reports the forecasting abilities of different methods: standard ordinary least squares (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the ratio of the root mean-squared forecasting error of the model to that of the random walk without drift: Values smaller than unity denote that the model forecasted better than the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one with the lowest RMSE ratio (when significant). The forecasts were for 12 months and used the monetary model.

Table 6
12-month forecast. Monetary model sticky prices.

	First difference specification					ECM specification				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	0.90*	0.83***	0.81***	0.75***	0.63***	0.76***	0.65***	0.69***	0.59***	0.32***
AT	1.04	1.06	0.87	0.75	0.23***	0.50**	0.40***	0.39***	0.52***	0.19***
BE	1.87	1.01	0.80*	0.84	0.41***	1.29	0.40***	0.20***	0.53***	0.19***
CA	0.99	1.04	1.04	0.88***	0.68***	0.74***	0.87***	0.63***	0.51***	0.33***
DK	1.01	1.00	0.85*	0.85**	0.51***	0.75***	0.44***	0.27***	0.57***	0.24***
FI	1.13	0.96	0.90*	0.85***	0.81**	0.64***	0.45***	0.40***	0.48***	0.30***
FR	1.12	1.00	0.67***	0.88**	0.52***	0.70***	0.97	0.64***	0.43***	0.23***
DE	1.35	0.89**	0.80***	0.88***	0.34***	0.99	0.62***	0.24***	0.39***	0.17***
GR	0.71**	0.55***	0.50***	0.56***	0.30***	0.63***	0.72***	0.75***	0.54***	0.19***
IE	1.46	1.42	1.27	1.23	0.82**	1.19	1.00	0.45***	0.73***	0.29***
IT	1.14	1.21	1.01	0.92*	0.62***	0.91	0.52***	0.35***	0.58***	0.25***
JP	0.90***	0.75***	0.66***	0.63***	0.35***	0.92*	0.46***	0.24***	0.38***	0.20***
NL	1.48	1.25	0.87	1.03	0.39***	0.37***	0.54***	0.42***	0.47***	0.19***
NZ	0.68***	0.70***	0.58***	0.65***	0.46***	0.99	0.55***	0.23***	0.51***	0.26***
PT	1.23	0.88***	0.58**	0.66***	0.28***	0.86*	0.70***	0.71***	0.53***	0.19***
ES	1.79	1.37	1.35	1.36	0.65***	0.91	0.63***	0.81***	0.54***	0.32***
SE	1.14	1.01	1.00	0.86***	0.70***	1.13	0.77***	0.68***	0.59***	0.24***
CH	1.08	0.88	1.14	0.47**	0.63	1.11	0.88	0.56*	0.32***	0.35***
GB	1.08	0.92**	0.81***	0.86***	0.83**	0.82**	0.72***	0.65***	0.54***	0.40***

The table reports the forecasting abilities of different methods: standard ordinary least squares (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the ratio of the root mean-squared forecasting error of the model to that of the random walk without drift: Values smaller than unity denote that the model forecasted better than the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one with the lowest RMSE ratio (when significant). The forecasts were for 12 months and used the monetary model, with sticky prices.

statistical tools.¹⁸ Specifically, we use two approaches. First, we calculate the ratio of the root mean-squared error (RMSE) of the model to that of a random

walk.¹⁹ A value smaller than one indicates that the model delivers better forecasting performance than a random walk. Inference is based on the Diebold and Mariano

¹⁸ Given the objective of the study, we do not consider an economic evaluation of predicting the exchange rate using, for example, asset allocation approaches as in Della Corte, Sarno, and Tsiakas (2009), Della Corte and Tsiakas (2012), Li et al. (2015).

¹⁹ The RMSE criterion is by far the most widely used approach in the literature. Alternatively, the mean absolute percentage error (MAPE) can be used although its interpretation is less clear in the presence of log variables, as in our case.

Table 7

12-month forecast. Monetary model sticky prices, UIP deviations.

	First difference specification					ECM specification				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	0.94	0.84**	0.77***	0.74***	0.55***	0.78***	0.63***	0.69***	0.59***	0.36***
AT	1.06	1.24	0.74	0.74	0.25***	0.52***	0.36***	0.34***	0.51***	0.19***
BE	2.12	1.08	0.90	0.87	0.36***	1.60	0.36***	0.22***	0.57***	0.23***
CA	0.92	1.05	1.04	0.82***	0.63***	0.72***	0.89***	0.72***	0.51***	0.38***
DK	0.73**	0.55***	0.39***	0.64***	0.33***	0.70**	0.69***	0.61***	0.42***	0.16***
FI	0.73**	0.82***	0.53***	0.75***	0.58***	0.41***	0.33***	0.18***	0.53***	0.40***
FR	1.16	1.01	0.65***	0.87**	0.44***	0.74***	0.95	0.57***	0.45***	0.32***
DE	1.44	0.90**	0.83***	0.87***	0.34***	0.93	0.48***	0.25***	0.42***	0.23***
IE	1.94	1.46	1.39	1.28	0.91	1.35	1.17	1.78	0.64***	0.50***
IT	0.86	1.31	0.78	0.72**	0.51***	0.56***	0.40***	0.30***	0.59***	0.42***
JP	0.75***	0.61***	0.49***	0.58***	0.31***	0.60***	0.43***	0.24***	0.37***	0.21***
NL	1.19	1.37	0.80	1.00	0.40***	0.35***	0.10***	0.40***	0.45***	0.19***
NZ	0.64***	0.62***	0.51***	0.64***	0.45***	0.71***	0.44***	0.29***	0.53***	0.26***
PT	1.05	1.27	0.75	0.81**	0.32***	0.62***	0.36***	0.21***	0.59***	0.18***
ES	1.86	1.24	1.32	1.41	0.67***	0.87*	1.18	0.90**	0.59***	0.36***
SE	1.17	1.08	1.00	0.92*	0.58***	1.19	5.32	51.04	0.55***	0.33***
CH	1.16	0.81	0.62	0.44**	0.43***	1.00	0.30***	0.85	0.62**	0.37***
GB	1.13	0.87**	0.77***	0.78***	0.70***	0.89*	0.70***	0.44***	0.53***	0.42***

The table reports the forecasting abilities of different methods: standard ordinary least squares (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the ratio of the root mean-squared forecasting error of the model to that of the random walk without drift: Values smaller than unity denote that the model forecasted better than the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one with the lowest RMSE ratio (when significant). The forecasts were for 12 months, and used the monetary model, with sticky prices and UIP deviations.

statistic (Diebold & Mariano, 1995). Given the full sample of observables $\{\Delta_k e_{t+k}, X_t\}_{t=1}^T$, where k is the size of the forecasting horizon and T the total size of the sample, we divide the sample between an in-sample portion with observations from 1 to L , and an out of sample portion with observations from $L + k$ to $T + k$. The size of the out-of-sample portion is therefore $M = T - L + 1$. We applied the rolling window forecasting scheme, where the parameters were re-estimated over time using the L most recent observations; in our case, the estimation window size $M = 120$ periods (10 years). This produced a sequence of M k forward-forecast errors that were used to compute the RMSE statistic.

Although the RMSE criterion is useful for assessing whether models can forecast the magnitude of movements in the exchange rate, an alternative approach focuses on whether the models can predict the *direction of change* of the exchange rate. Given the financial nature of the exchange rate, the direction of change is a useful signal for designing meaningful and profitable investment strategies. We therefore use the direction of change statistic, which is essentially the percentage of mistakes that models make in directional forecasts. A value below 50% represents better forecasting performance against the null of the random walk. In this case, inference is based on the Diebold and Mariano statistic.

Robustness over time is assessed through the [Giacomini and Rossi \(2010\)](#) fluctuation test that measures the relative local forecasting performance of the model, as opposed to that of random walk, and investigates its stability over time.

4.4. Data

Our analysis focused on post-Bretton Woods exchange rates for advanced economies. We used non-seasonally

Table 8

12-month forecast. Taylor rule model.

	ECM specification				
	REG	SP	SPI	RF	SVM
AU	0.55***	0.76***	0.66***	0.58***	0.45***
AT	0.86*	0.42***	0.27***	0.59***	0.30***
BE	0.81**	0.40***	0.38***	0.64***	0.34***
CA	0.89*	0.95	0.91***	0.58***	0.46***
DK	0.75***	0.77***	0.80***	0.62***	0.49***
FI	0.69***	0.52***	0.31***	0.59***	0.41***
FR	0.72***	0.58***	0.61***	0.54***	0.30***
DE	0.90	0.43***	0.32***	0.57***	0.29***
IE	0.87	0.44***	0.48***	0.76***	0.52***
IT	0.66***	1.01	1.01	0.48***	0.35***
JP	0.61***	0.47***	0.31***	0.52***	0.28***
NL	0.74***	0.53***	0.44***	0.59***	0.46***
NZ	0.30***	0.15***	0.20***	0.47***	0.32***
PT	0.49***	0.22***	0.25***	0.54***	0.27***
ES	0.74***	1.20	1.20	0.66***	0.28***
SE	0.89*	0.78***	0.78***	0.58***	0.46***
CH	2.49	1.03	0.75	1.16	0.33***
GB	0.79***	0.76***	0.75***	0.68***	0.45***

The table reports the forecasting abilities of different methods: standard ordinary least squares (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the ratio of the root mean-squared forecast error of the model to that of random walk without drift: values smaller than unity denote that the model forecasted better than the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one with the lowest RMSE ratio (when significant). The forecasts were for 12 months and used the Taylor rule model.

adjusted data, implementing our own seasonal adjustment using only backward information with a lagged MA(12) filter. Seasonal adjustment was implemented for prices, CPI (consumer price index), money, and output (industrial production index). The variables are end-of-period values derived from standard sources (mainly, IMF

Table 9

1-month forecast. Sign prediction. Monetary model and monetary model with sticky prices, UIP deviations. First difference specification.

	Monetary model					Monetary model, sticky prices, UIP deviations				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	0.51	0.49	0.49	0.47	0.53	0.46*	0.52	0.52	0.48	0.48
AT	0.51	0.41	0.41	0.41	0.49	0.51	0.41	0.41	0.46	0.41
BE	0.41**	0.43	0.43	0.47	0.49	0.38**	0.43	0.39**	0.42	0.39**
CA	0.51	0.51	0.51	0.50	0.54	0.50	0.51	0.51	0.52	0.53
DK	0.50	0.49	0.49	0.54	0.53	0.49	0.54	0.54	0.56	0.53
FI	0.54	0.51	0.51	0.53	0.53	0.54	0.53	0.53	0.56	0.53
FR	0.47	0.42*	0.49	0.46	0.47	0.43*	0.47	0.49	0.43*	0.41**
DE	0.51	0.48	0.48	0.52	0.53	0.52	0.47	0.50	0.48	0.50
GR	0.52	0.53	0.53	0.57	0.51	n.a.	n.a.	n.a.	n.a.	n.a.
IE	0.51	0.46	0.46	0.51	0.50	0.44	0.50	0.49	0.53	0.48
IT	0.41	0.38**	0.40*	0.41*	0.41*	0.45	0.37**	0.38**	0.51	0.49
JP	0.54	0.48	0.48	0.49	0.50	0.51	0.51	0.51	0.46**	0.47*
NL	0.42	0.42	0.42	0.39	0.47	0.45	0.42	0.42	0.37*	0.47
NZ	0.48	0.42**	0.42**	0.42**	0.40**	0.43*	0.42**	0.42**	0.45	0.41**
PT	0.54	0.53	0.54	0.50	0.49	0.40*	0.46	0.44	0.54	0.54
ES	0.46	0.52	0.51	0.53	0.48	0.47	0.54	0.52	0.47	0.46
SE	0.52	0.49	0.49	0.49	0.51	0.46*	0.49	0.49	0.48	0.48
CH	0.50	0.79	0.79	0.54	0.50	0.46	0.54	0.50	0.50	0.50
GB	0.45*	0.50	0.50	0.51	0.53	0.48	0.50	0.50	0.52	0.47

The table reports the forecasting abilities of different methods: logistic regression (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the proportion of mistakes in predicting the direction of the rate of movement of dollar exchange; values below 1/2 denote better forecasting ability with respect to the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one with the lowest value (when significant). The forecasts were for one month ahead, and used the monetary model (panel above) and monetary model with sticky prices and UIP deviations (panel below).

Table 10

1-month forecast. Sign prediction. Monetary model with sticky prices.

	First difference specification					ECM specification				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	0.45**	0.49	0.49	0.49	0.50	0.48	0.51	0.51	0.54	0.49
AT	0.49	0.41	0.41	0.35**	0.38	0.53	0.47	0.47	0.39*	0.45
BE	0.43	0.43	0.43	0.46	0.39	0.49	0.44	0.44	0.41*	0.44
CA	0.51	0.50	0.50	0.53	0.54	0.53	0.50	0.50	0.48	0.49
DK	0.49	0.47	0.47	0.50	0.50	0.48	0.51	0.52	0.52	0.49
FI	0.51	0.53	0.51	0.56	0.53	0.47	0.48	0.44*	0.50	0.46
FR	0.42	0.44	0.49	0.52	0.47	0.46	0.47	0.51	0.56	0.45
DE	0.52	0.48	0.48	0.45	0.48	0.55	0.53	0.51	0.50	0.47
GR	0.54	0.51	0.51	0.53	0.44	0.49	0.52	0.49	0.46	0.46
IE	0.50	0.46	0.46	0.52	0.48	0.48	0.49	0.49	0.49	0.42**
IT	0.41*	0.52	0.41*	0.55	0.44	0.58	0.54	0.65	0.48	0.48
JP	0.53	0.50	0.49	0.48	0.46*	0.48	0.49	0.50	0.51	0.47*
NL	0.42	0.42	0.42	0.47	0.42	0.54	0.46	0.46	0.46	0.50
NZ	0.43*	0.42**	0.42**	0.38***	0.43*	0.50	0.39***	0.39***	0.40**	0.38***
PT	0.42	0.51	0.47	0.55	0.49	0.56	0.53	0.53	0.55	0.55
ES	0.46	0.51	0.53	0.49	0.44**	0.44**	0.42**	0.51	0.46	0.41**
SE	0.47	0.49	0.49	0.50	0.49	0.48	0.51	0.53	0.50	0.47*
CH	0.46	0.54	0.54	0.58	0.46	0.50	0.47	0.47	0.58	0.64
GB	0.48	0.50	0.50	0.49	0.50	0.54	0.50	0.50	0.50	0.51

The table reports the forecasting abilities of different methods: logistic regression (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), and SVMs (SVM). For each method is displayed the proportion of mistakes in predicting the direction of the rate of movement of dollar exchange; values below 1/2 denote better forecasting ability with respect to the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one that yielded the lowest value (when significant). The forecasts were for one month, and used the monetary model with sticky prices.

International Financial Statistics (IFS)). All the series were downloaded from Datastream, and covered the period from April 1973 (end of the Bretton Woods agreement) to December 2017. For some countries, the sample was restricted due to data availability; in particular, for Eurozone member countries, the estimation ended in December 1998. The detailed data sources were as follows (whenever the Datastream code is reported, CC represents the country code):

1. Exchange rates: national currency per US dollar, Datastream, IMF IFS, (CCI.AE.)
2. Money: money stock M1, Datastream, IMF IFS, (code CCM1....A)
3. Output: industrial production index, Datastream, IMF IFS (code CCI66...F)
4. Interest rate: money market rate, Datastream, IMF IFS (code CC60B..ZF...)

Table 11
1-month forecast. Sign prediction. Taylor rule model.

	ECM specification				
	REG	SP	SPI	RF	SVM
AU	0.46*	0.52	0.55	0.54	0.51
AT	0.58	0.51	0.56	0.43**	0.44*
BE	0.56	0.49	0.45*	0.45*	0.44**
CA	0.51	0.51	0.51	0.52	0.49
DK	0.51	0.52	0.54	0.50	0.45*
FI	0.49	0.54	0.51	0.48	0.49
FR	0.48	0.50	0.50	0.44**	0.44**
DE	0.54	0.50	0.44*	0.44**	0.46
IE	0.51	0.52	0.46	0.43*	0.50
IT	0.46	0.53	0.47	0.38***	0.44**
JP	0.48	0.49	0.49	0.51	0.49
NL	0.54	0.47	0.52	0.47	0.47
NZ	0.42**	0.40**	0.40**	0.36***	0.35***
PT	0.44	0.47	0.47	0.47	0.42
ES	0.45	0.53	0.49	0.45	0.40***
SE	0.51	0.54	0.53	0.46*	0.50
CH	0.61	0.36	0.36*	0.67	0.72
GB	0.53	0.50	0.50	0.53	0.52

The table reports the forecasting abilities of different methods: logistic regression (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), and SVMs (SVM). For each method is displayed the proportion of mistakes in predicting the direction of movement of dollar exchange; values below 1/2 denote better forecasting ability with respect to the random walk benchmark. *, **, and ***, denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., the one with the lowest value (when significant). The forecasts were for one month and used the Taylor rule model.

- Prices: consumer price index, Datastream, national statistical offices (code CCONPRCF)

5. Results

5.1. Forecasting performance

The first analysis implemented was a standard exercise to assess the performance of the different exchange rate models. For each model, we produced one-month and 12-months forecasts using different methods; we assessed performance using the RMSE and the direction of change.

Tables 1–14 present the detailed results. Each table assessed the performance of an exchange rate model (1. monetary model, 2. monetary model with sticky prices, 3. monetary model with sticky prices and deviations from UIP, and 4. Taylor rule model). For each model, we produced one-month and 12-month forecasts using different methods: standard ordinary least squares (REG),²⁰ elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), and the SVMs (SVM). Specifically, Tables 1–8 assess performance using RMSE: a value below one denotes a model that performs better than random walk. Tables 9–14 assess performance based on the direction of change. To keep interpretations across tables consistent, we measured mistakes in predicting the direction of movement of the exchange rate. Therefore, a value

below 0.5 indicated that the model had outperformed the random walk benchmark. In each case, we implemented a model specification with first differences as in Eq. (1), and one with error correction as in Eq. (3). The bolded values identify the model that delivered the best performance (when significant) for each country.

The results can be summarized as follows:

First, standard exchange rate models, irrespective of the estimation method, even with SL tools, failed to outperform the random walk in the very short run (one month ahead). This was true for both standard forecasts and forecasts of the direction of change, confirming the unpredictability of asset prices in very short horizons.

Second, the performance of standard regression models improved significantly for long-run forecasts. This result confirms not only the general findings of the literature, but also the standard practice of market makers and practitioners, who tend to use fundamentals-based models for long-run forecasts while relying on technical analysis for the short run (Cheung & Chinn, 2001; Dick & Menkhoff, 2013).

Third, in a comparison of exchange rate models, sticky prices models (with or without deviations from UIP) exhibited the best predictive performance, although the Taylor rule model also performed well in the long run.

Fourth, SL methods consistently outperformed standard regression models in forecasting the exchange rate, both in the short and long runs. Although in the short run, this was not enough to outperform random walk, in the long run, SL tools were clearly better irrespective of the model, country, statistic (magnitude or direction of change), and model specification (first difference or ECM).²¹ The magnitude of the statistics is worth noting: The ratio of RMSEs was mostly below 0.5 and often below 0.3, in some cases showing the remarkable predictive abilities of SL tools.

Overall, more flexible tools allowing for non-linearities and interaction effects seemed to perform better, especially with more complex models. This is clear when comparing results using the RMSE with those using the direction of change. Correctly predicting the direction of change of the exchange rate in the long run is certainly easier than predicting the magnitude of this change; moreover, as a dichotomous choice, non-linearities matter less. As a result, in directional forecasts, SL tools outperformed standard regression, but by less than they did when the task was to predict the magnitude of the movement of exchange rate. Finally, there was clearly a best performer among SL tools: The SVM won in 85% of the cases.

Having established these stylized facts emerging from our dataset, without loss of generality, we proceed by focusing on an SVM as reference SL tool, and the more general monetary model with sticky prices and UIP deviations as the benchmark model (ECM specification).

²⁰ Given the dichotomous nature of the response variable in case of sign prediction, we used a logit instead of standard OLS.

²¹ A formal test of cross-model comparison supports the superiority of SL tools with respect to regression. The results are available upon request.

Table 12

12-month forecast. Sign prediction. Monetary model and monetary model with sticky prices and UIP deviations. First difference specification.

	Monetary model					Monetary model, sticky prices, UIP deviations				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	0.40***	0.41***	0.40***	0.37***	0.32***	0.41***	0.38***	0.37***	0.36***	0.22***
AT	0.24***	0.44	0.44	0.32*	0.24***	0.40	0.44	0.12***	0.32*	0.12***
BE	0.56	0.44	0.32***	0.40*	0.21***	0.58	0.38	0.32***	0.23***	0.13***
CA	0.49	0.54	0.53	0.40***	0.36***	0.44***	0.56	0.58	0.36***	0.24***
DK	0.38***	0.48	0.38***	0.39***	0.32***	0.42	0.33**	0.40	0.38*	0.36**
FI	0.48	0.58	0.58	0.41**	0.36***	0.39**	0.24***	0.23***	0.35***	0.22***
FR	0.27***	0.38**	0.20***	0.24***	0.22***	0.30***	0.16***	0.19***	0.24***	0.17***
DE	0.62	0.59	0.61	0.52	0.34***	0.58	0.46	0.34***	0.45	0.13***
GR	0.22***	0.22***	0.22***	0.22***	0.16***	n.a.	n.a.	n.a.	n.a.	n.a.
IE	0.48	0.53	0.59	0.55	0.51	0.55	0.65	0.48	0.57	0.38**
IT	0.36*	0.21***	0.28***	0.21***	0.23***	0.48	0.28***	0.34***	0.23***	0.18***
JP	0.36***	0.32***	0.30***	0.27***	0.25***	0.26***	0.22***	0.18***	0.22***	0.11***
NL	0.27**	0.88	0.27**	0.38	0.19***	0.38	0.42	0.15***	0.42	0.15***
NZ	0.36***	0.34***	0.31***	0.32***	0.30***	0.36***	0.37***	0.30***	0.29***	0.19***
PT	0.35**	0.48	0.52	0.42	0.40*	0.36*	0.42	0.28***	0.14***	0.14***
ES	0.45	0.42**	0.43**	0.32***	0.32***	0.49	0.44	0.47	0.40***	0.23***
SE	0.39***	0.39***	0.29***	0.30***	0.27***	0.41***	0.33***	0.32***	0.30***	0.19***
CH	0.08***	0.08***	0.25*	0.17**	0.17**	0.08***	0.50	0.58	0.33	0.17**
GB	0.42**	0.37***	0.36***	0.30**	0.30***	0.39***	0.37***	0.35***	0.30***	0.24***

The table reports the forecasting abilities of different methods: logistic regression (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the proportion of mistakes in predicting the direction of the rate of movement of dollar exchange; values below 1/2 denote better forecasting ability with respect to the random walk benchmark. *, **, and *** denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., that displays the lowest value (when significant). The forecasts were for 12 months, and used the monetary model (panel above) and monetary model with sticky prices and UIP deviations (panel below).

Table 13

12-month forecast. Sign prediction. Monetary model with sticky prices.

	First difference specification					ECM specification				
	REG	SP	SPI	RF	SVM	REG	SP	SPI	RF	SVM
AU	0.39***	0.37***	0.35***	0.34***	0.27***	0.33***	0.24***	0.19***	0.21***	0.19***
AT	0.44	0.44	0.36	0.36	0.12***	0.24***	0.03***	0.05***	0.14***	0.05***
BE	0.61	0.37**	0.13***	0.21***	0.15***	0.38**	0.08***	0.08***	0.22***	0.14***
CA	0.49	0.58	0.44**	0.41***	0.29***	0.31***	0.35***	0.17***	0.17***	0.14***
DK	0.33***	0.38***	0.38***	0.38***	0.29***	0.30***	0.26***	0.20***	0.22***	0.16***
FI	0.46	0.56	0.39***	0.33***	0.28***	0.32***	0.14***	0.10***	0.18***	0.12***
FR	0.37**	0.26***	0.23***	0.24***	0.23***	0.36***	0.23***	0.17***	0.17***	0.16***
DE	0.61	0.46	0.37***	0.40**	0.17***	0.39***	0.27***	0.27***	0.16***	0.11***
GR	0.22***	0.22***	0.16***	0.25***	0.15***	0.39**	0.19***	0.19***	0.15***	0.14***
IE	0.61	0.65	0.52	0.54	0.38***	0.37***	0.37***	0.27***	0.27***	0.18***
IT	0.38**	0.25***	0.36**	0.25***	0.20***	0.25***	0.22***	0.21***	0.18***	0.16***
JP	0.39***	0.26***	0.24***	0.22***	0.15***	0.32***	0.22***	0.20***	0.12***	0.10***
NL	0.46	0.88	0.12***	0.50	0.15***	0.21***	0.24***	0.11***	0.11***	0.05***
NZ	0.41**	0.37***	0.23***	0.27***	0.20***	0.49	0.31***	0.26***	0.13***	0.13***
PT	0.37**	0.40*	0.29***	0.16***	0.15***	0.27***	0.14***	0.12***	0.22***	0.09***
ES	0.46	0.46	0.45	0.37***	0.25***	0.28***	0.13***	0.14***	0.12***	0.17***
SE	0.40***	0.54	0.30***	0.29***	0.22***	0.35***	0.29***	0.23***	0.22***	0.13***
CH	0.08***	0.50	0.58	0.50	0.33	0.21***	0.17***	0.04***	0.13***	0.17***
GB	0.40***	0.38***	0.33***	0.33***	0.30***	0.32***	0.32***	0.30***	0.22***	0.17***

The table reports the forecasting abilities of different methods: logistic regression (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the proportion of mistakes in predicting the direction of the rate of movement of dollar exchange; values below 1/2 denote better forecasting ability with respect to the random walk benchmark. *, **, and *** denote significance levels at 10%, 5%, and 1% of the p-values for the Diebold–Mariano test. The bolded values identify the best model, i.e., that displays the lowest value (when significant). The forecasts were for 12 months and used the monetary model with sticky prices.

5.2. Model and specification comparison

Given the consistently good forecasting performance of SL tools, we can exploit them to compare different exchange rate models. Focusing on 12-month forecasts,

we use SVMs to compare model performance across specifications.

Fig. 2 displays the results for individual countries, comparing the difference specification with the ECM (currencies). Overall, the ECM specification delivered lower RMSE

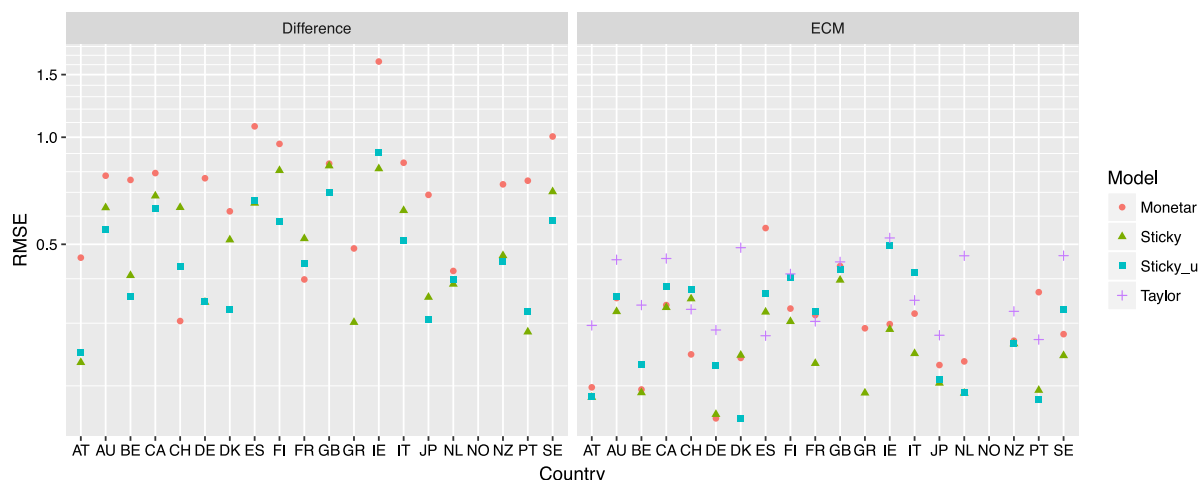


Fig. 2. Model performance comparison. svm out of sample, rolling estimates.

Table 14
12-month forecast. Sign prediction. Taylor rule model.

	ECM specification				
	REG	SP	SPI	RF	SVM
AU	0.25***	0.24***	0.18***	0.23***	0.17***
AT	0.42**	0.19***	0.17***	0.24***	0.13***
BE	0.39***	0.21***	0.21***	0.19***	0.13***
CA	0.30***	0.24***	0.20***	0.22***	0.15***
DK	0.33***	0.18***	0.21***	0.23***	0.16***
FI	0.37***	0.30***	0.26***	0.30***	0.13***
FR	0.41***	0.21***	0.16***	0.19***	0.14***
DE	0.39***	0.30***	0.2***	0.25***	0.13***
IE	0.37***	0.27***	0.26***	0.20***	0.23***
IT	0.31***	0.26***	0.25***	0.2***	0.13***
JP	0.22***	0.19***	0.11***	0.13***	0.10***
NL	0.38***	0.22***	0.22***	0.22***	0.16***
NZ	0.14***	0.08***	0.11***	0.18***	0.13***
PT	0.15***	0.09***	0.13***	0.15***	0.11***
ES	0.29***	0.30***	0.24***	0.23***	0.10***
SE	0.33***	0.29***	0.23***	0.22***	0.16***
CH	0.42	0.46	0.33*	0.50	0.21***
GB	0.34***	0.24***	0.23***	0.25***	0.17***

The table reports the forecasting abilities of different methods: standard ordinary least squares (REG), elastic net on splines (SP), elastic net on splines with interactions (SPI), random forest (RF), SVMs (SVM). For each method is displayed the proportion of mistakes in predicting the direction of the rate of movement of dollar exchange; values below 1/2 denote better forecasting ability than the random walk benchmark. *, **, and *** denote significance levels at 10%, 5%, and 1% of the p-values for relative statistical test. The bolded values identify the best model, i.e., the one with the lowest value (when significant). The forecasts were for 12 months, and used the Taylor rule model.

ratio values than the difference specification, with values consistently below 0.5. Looking at the different exchange rate models, despite some differences across countries, the sticky prices versions of the monetary model appear preferable.

Fig. 3 provides a more compact representation by showing the cross-country mean RMSE of predictions of the svm for different types of models and specifications. In general, sticky price specifications performed better both on the difference and the ECM specifications. This is consistent with the fact that at least in the long run,

deviations from the PPP, and the presence of an equilibrium relationship between the exchange rate and the fundamentals are relevant for predictive purposes.

5.3. Forecasting horizon

The previous section established that *sl* methods outperformed standard tools in forecasting the exchange rate 12 months ahead but failed to outperform the random walk on one-month forecasts. The natural question that arises relates to the horizon at which the fundamentals contain information that can be used by *sl* to forecast the exchange rate.

In Fig. 4, we show the performance of an svm at different forecasting horizons. The figure plots the RMSE ratio for each month for one- to 12-month forecasts. There was a striking similarity across countries. The performance of the svm improved almost monotonically with the length of the forecasting horizon up to 9–12 months, where it stabilized. Note that the svm model already exhibited good predictive ability at the three-month horizon.

We can therefore generalize the results of the previous section: With the exception of the very short run (one to two months), the *sl* methods consistently outperformed standard forecasting tools, irrespective of the country or horizon considered.

5.4. Forecast consistency across time

The results of the previous section were obtained using the entire forecasting period. However, the literature stresses that the choice of the forecasting horizon matters greatly; there are substantial changes over time in the predictive abilities of exchange rate models (Cheung et al., 2005; Giacomini & Rossi, 2010; Rogoff & Stavrageva, 2008).

To investigate this issue, Figs. 5–22 report the results of the Giacomini and Rossi (2010) fluctuation test. The test assesses variations in time in the models' relative forecasting performance by computing the rolling-window local relative RMSE for the model and the random walk

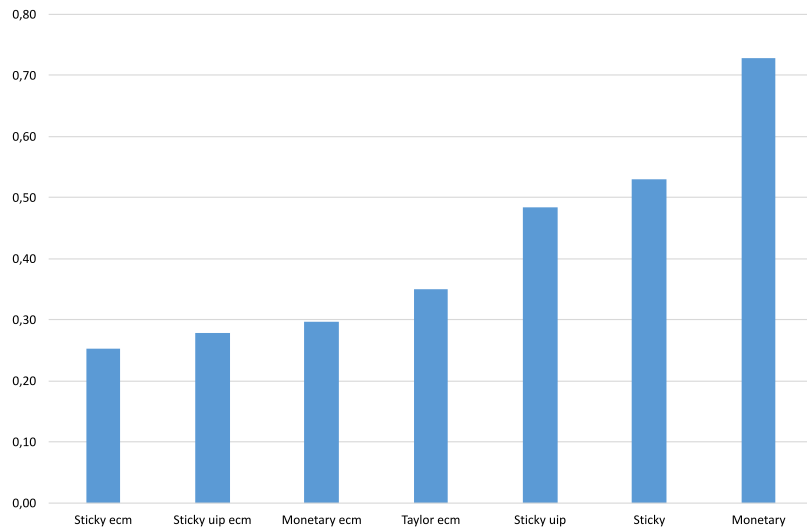


Fig. 3. Exchange rate models comparison. svm out of sample, rolling estimates. Mean RMSE across different currencies.

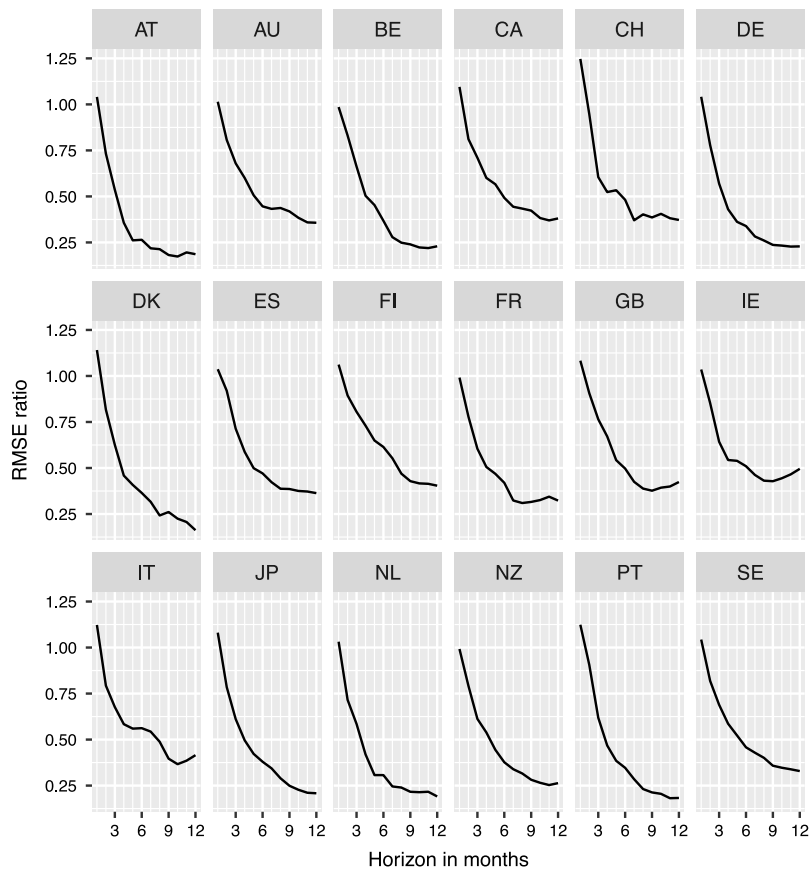


Fig. 4. Model performance comparison. Monetary model with sticky prices and uip deviations, svm, forecasts at different horizons (1–12 months ahead).

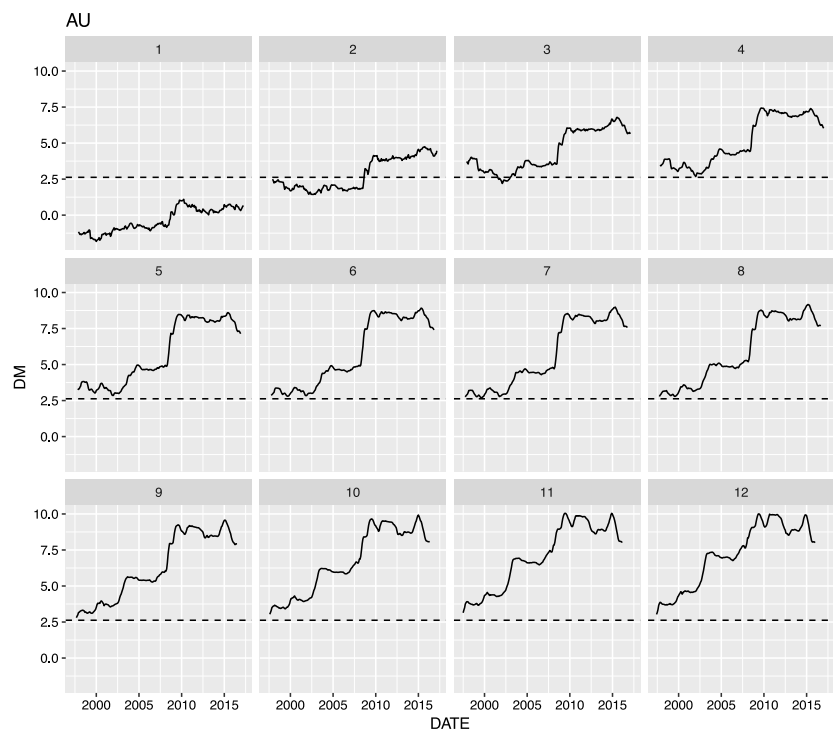


Fig. 5. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

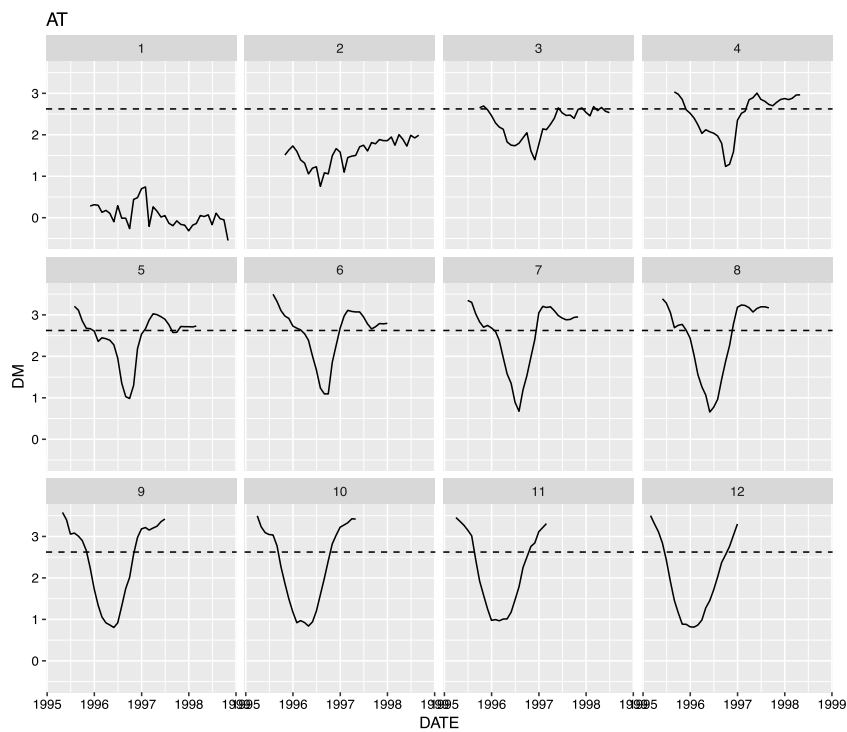


Fig. 6. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

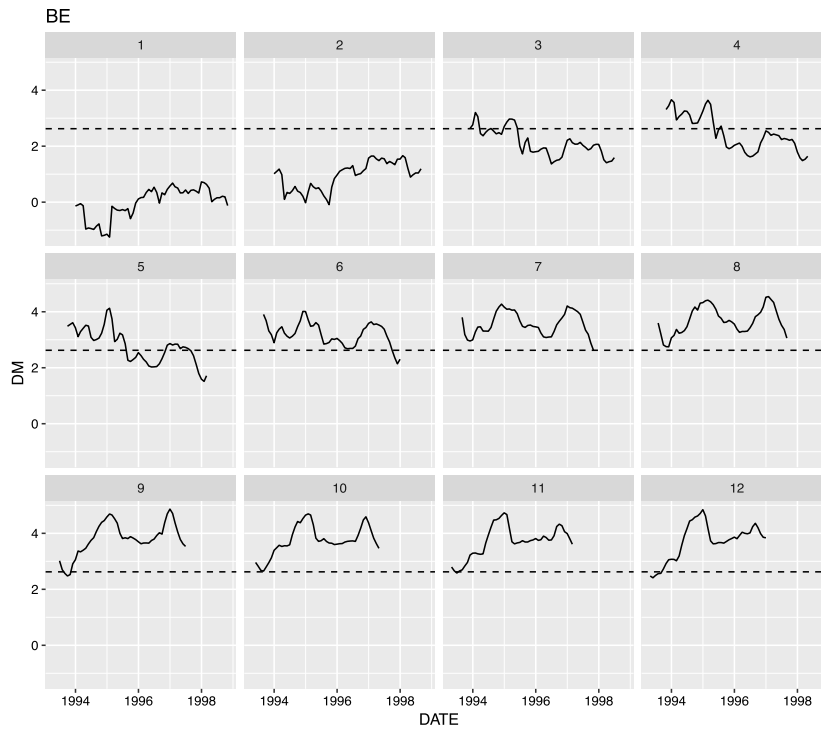


Fig. 7. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

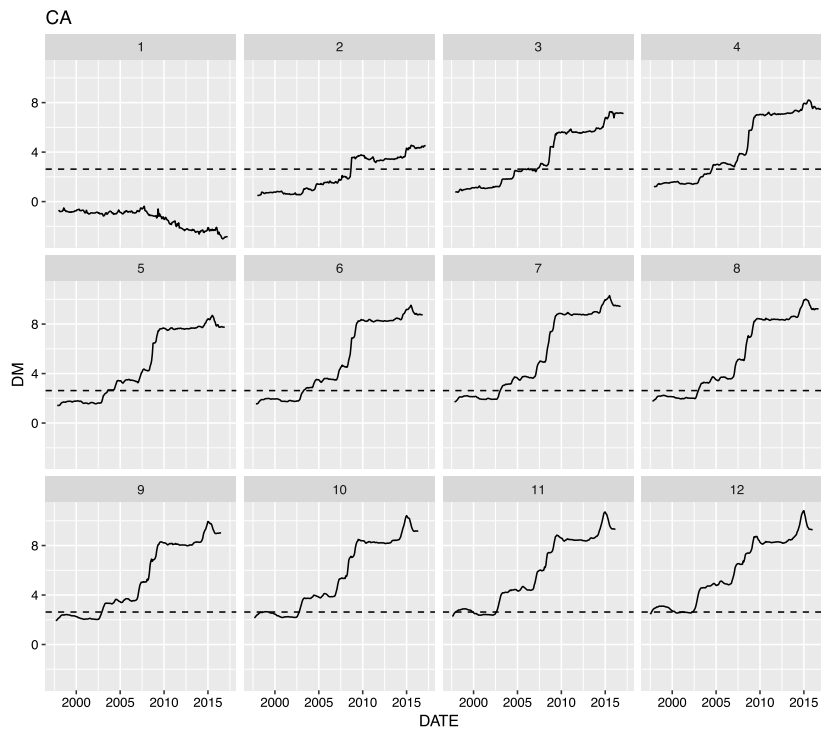


Fig. 8. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

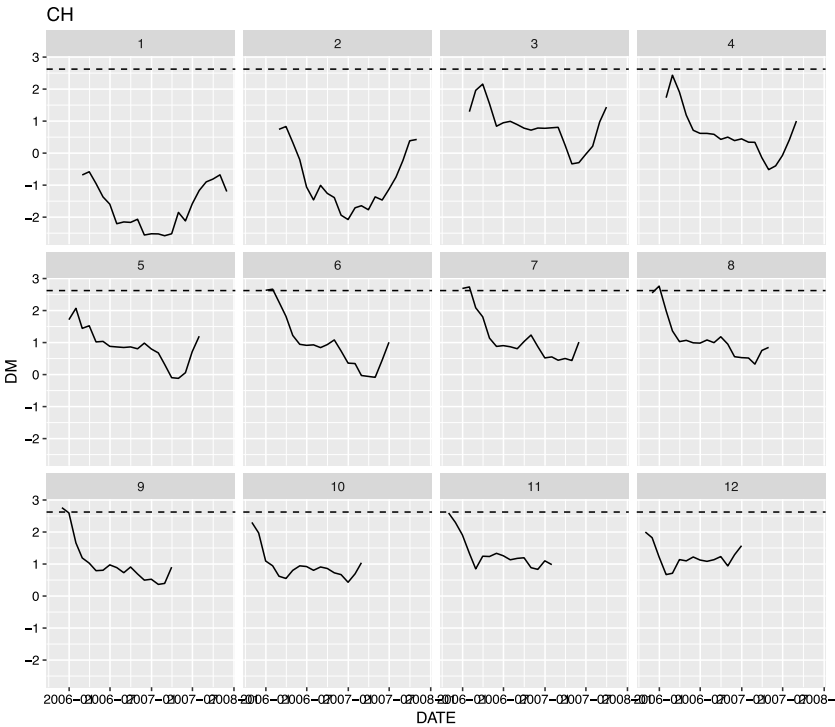


Fig. 9. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.



Fig. 10. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

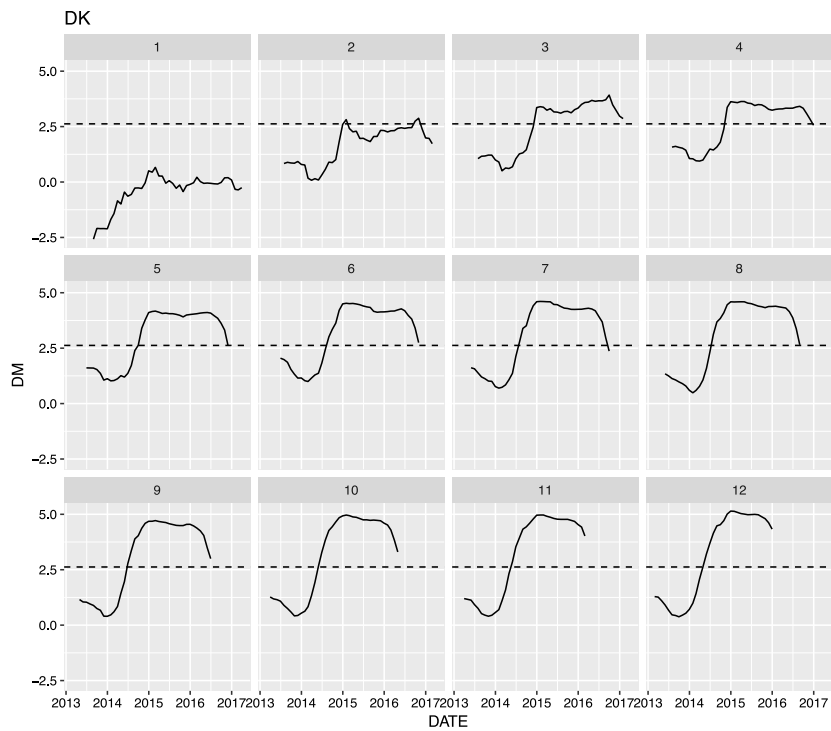


Fig. 11. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

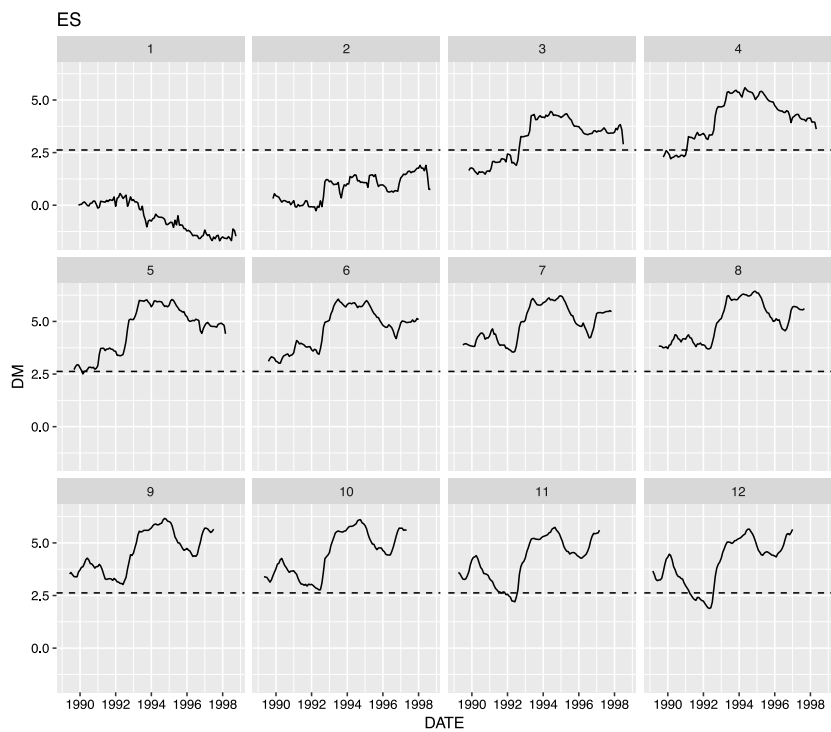


Fig. 12. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

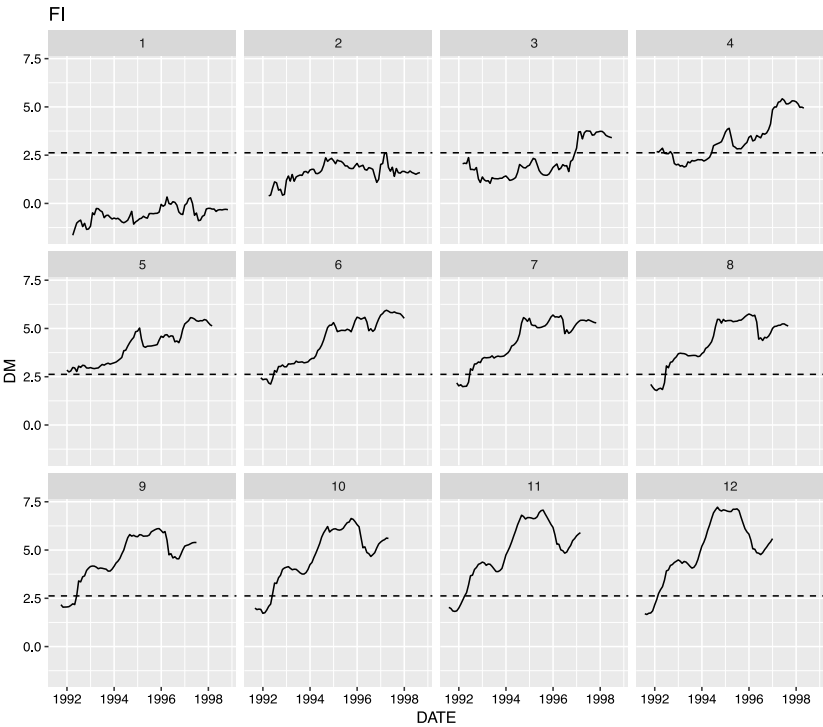


Fig. 13. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

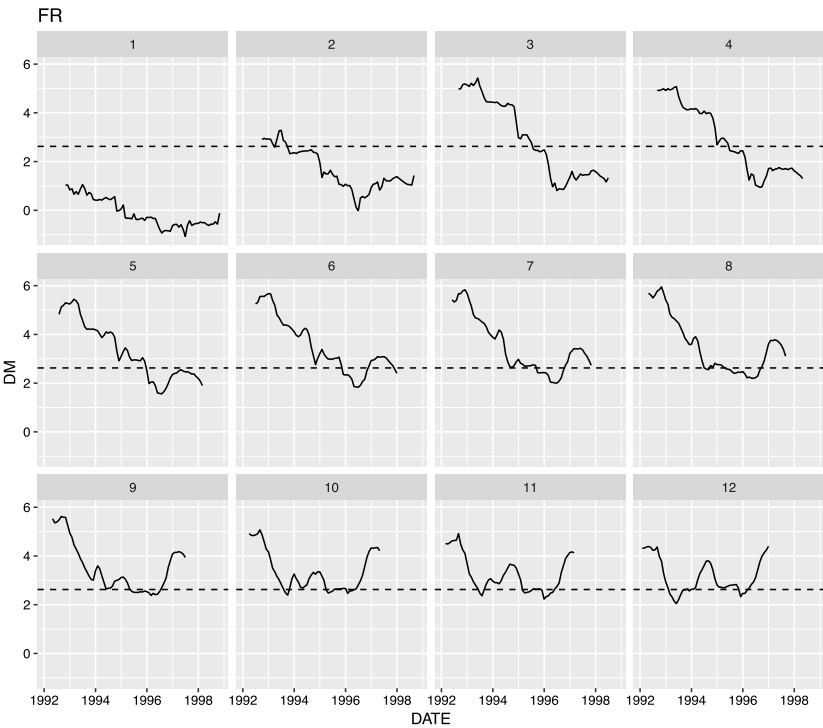


Fig. 14. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

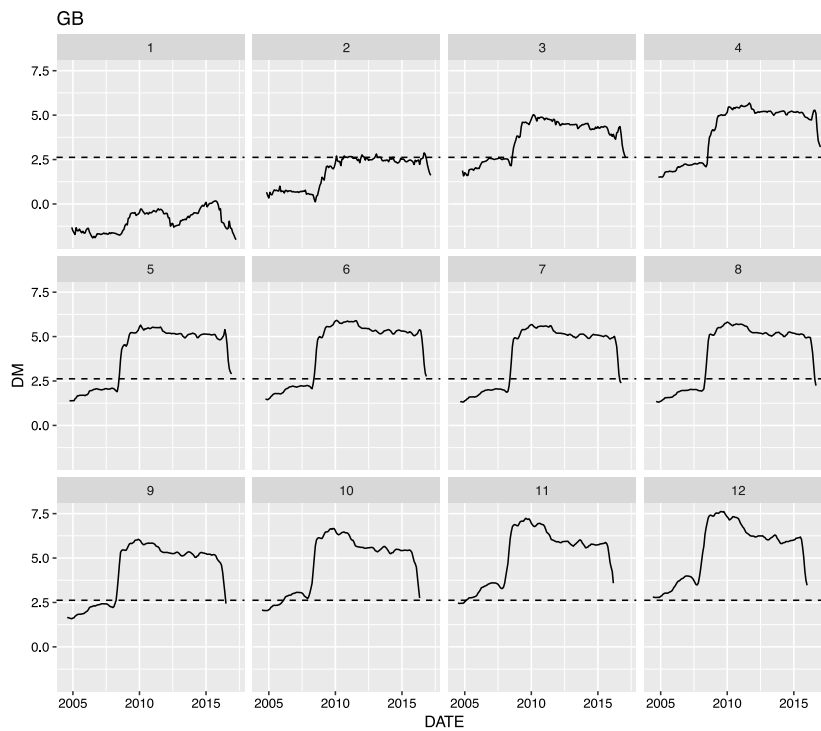


Fig. 15. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

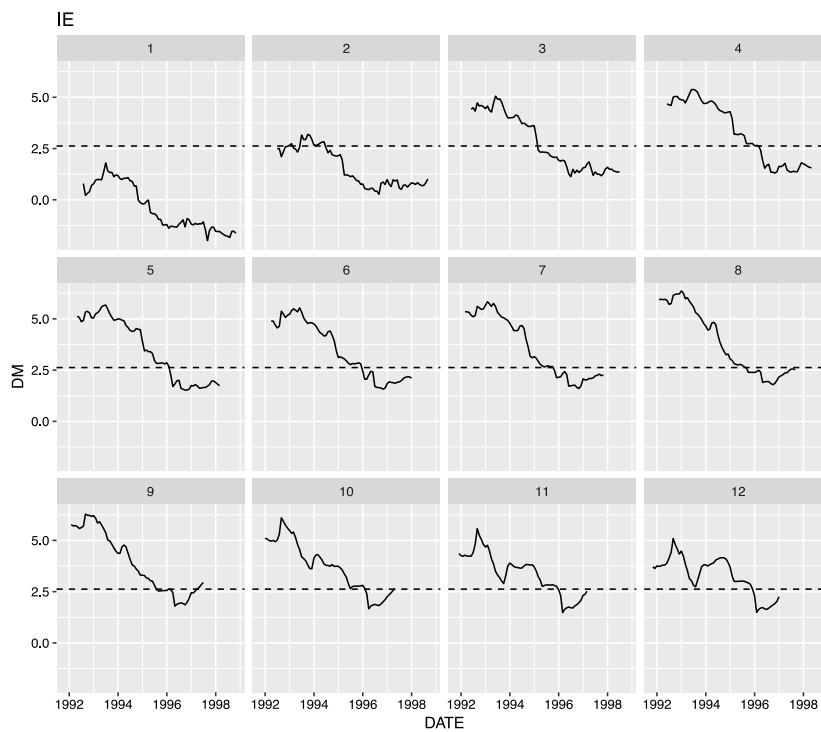


Fig. 16. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

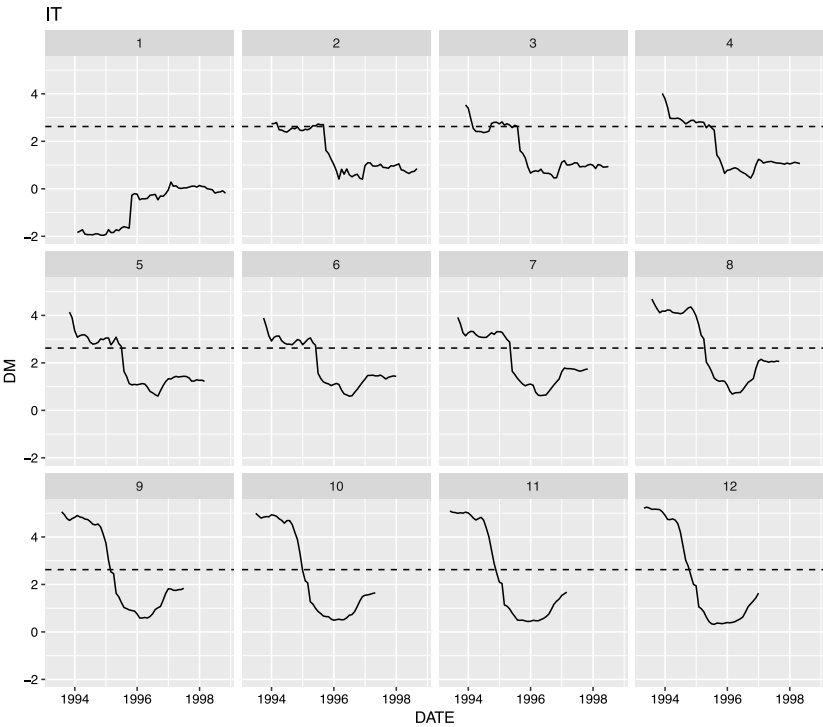


Fig. 17. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

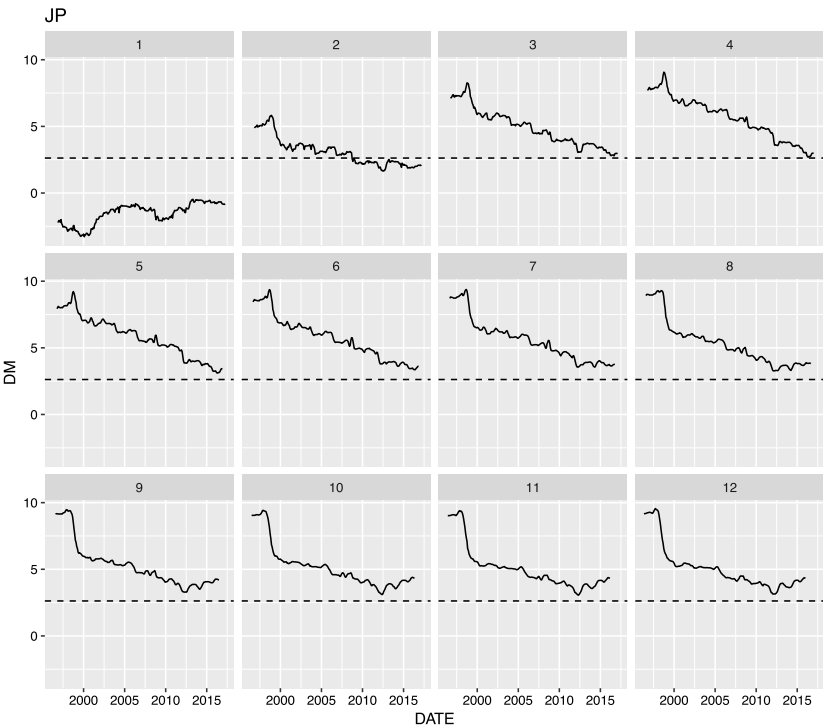


Fig. 18. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

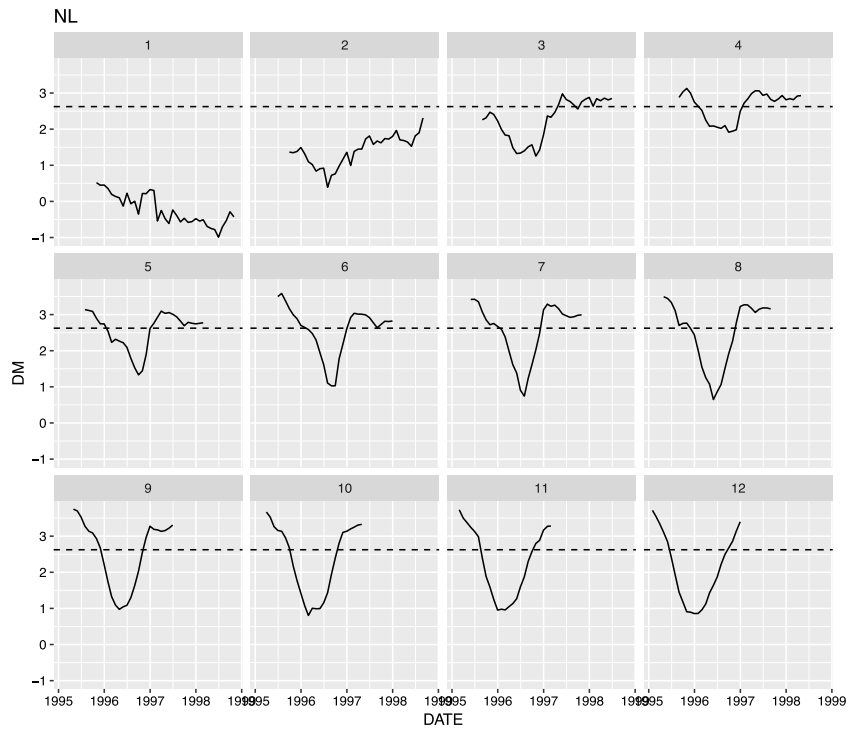


Fig. 19. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

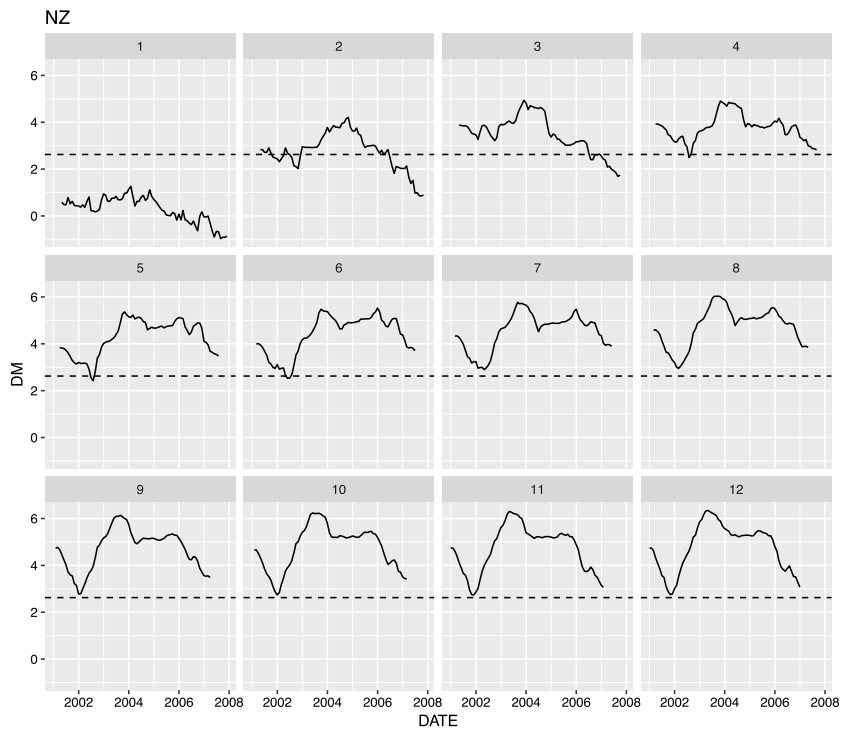


Fig. 20. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

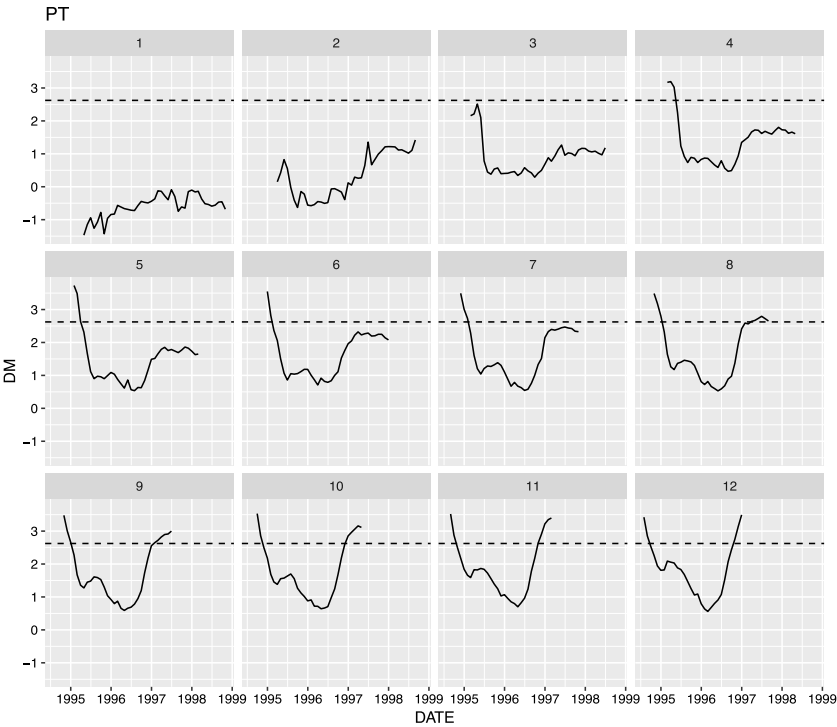


Fig. 21. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

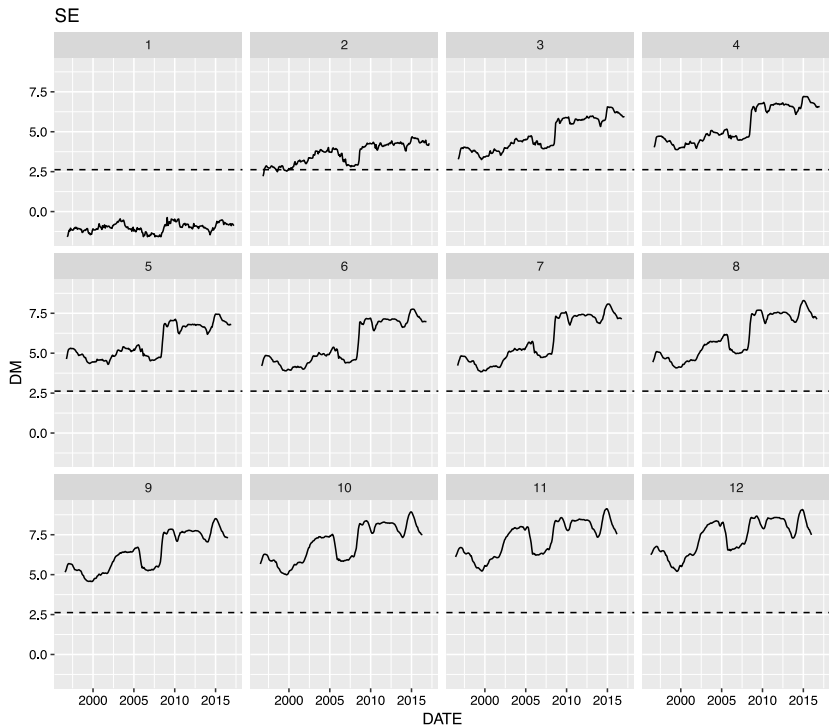


Fig. 22. Fluctuation test. Monetary model with sticky prices and UIP deviations, 1 to 12 months ahead forecasts, svm.

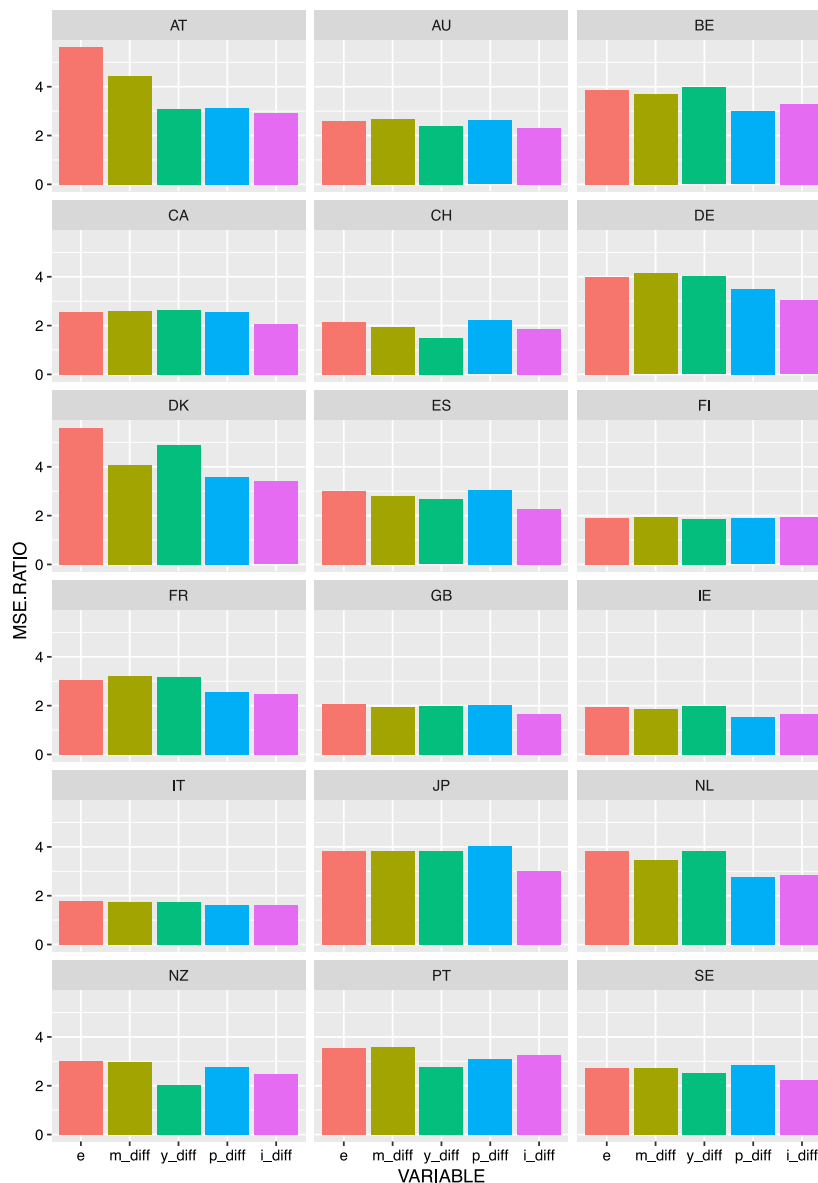


Fig. 23. Variable importance, out-of-sample 12-month forecasts, svm tool.

benchmark. The test's null hypothesis was that at each point in time, the two models had the same predictive performance.

The figures show the test statistics and critical values²² (dotted line). Values above the critical line implied that the model had better forecasting performance than the random walk. This was done for each country and each forecasting horizon from one to 12 months.

The figures confirm the results of the previous sections: With the exception of the very short run (one to

two months), the sl model did better than the random walk in different forecasting periods for most countries. In particular, it did so consistently for AU, BE, CA, DE, ES, FI, FR, GB, IE, JP, NZ, and SG, it occasionally for AT, DK, IT, and NL. It failed to pass the test only for CH and PT.

Thus, sl tools can produce forecasts that are robust against the choice of forecasting period. This is likely due to their flexibility—one of the major strength of sl techniques—which allows them to maximize their performance in the presence of environments with parameter instability.²³

²² We set the ratio of the size of the rolling window to the out-of-sample size at 0.4. The critical values were used for the one-sided test.

²³ Rossi (2006) showed that parameter instability is a relevant issue in exchange rate prediction.



Fig. 24. Partial dependence plots, 12-month forecasts, svm.

5.5. Variable importance

The results so far show that standard exchange rate models of the *sl* algorithm do have good predictive power. This means that the fundamentals contain information that can be exploited to consistently predict movements

of the exchange rate. The question that arises is whether some variables help predict movements in the exchange rate better than others. The previous section showed that the best model was the monetary model “augmented” by considering deviations in the PPP and UIP. This task is not easy in the context of statistical learning. On the contrary,

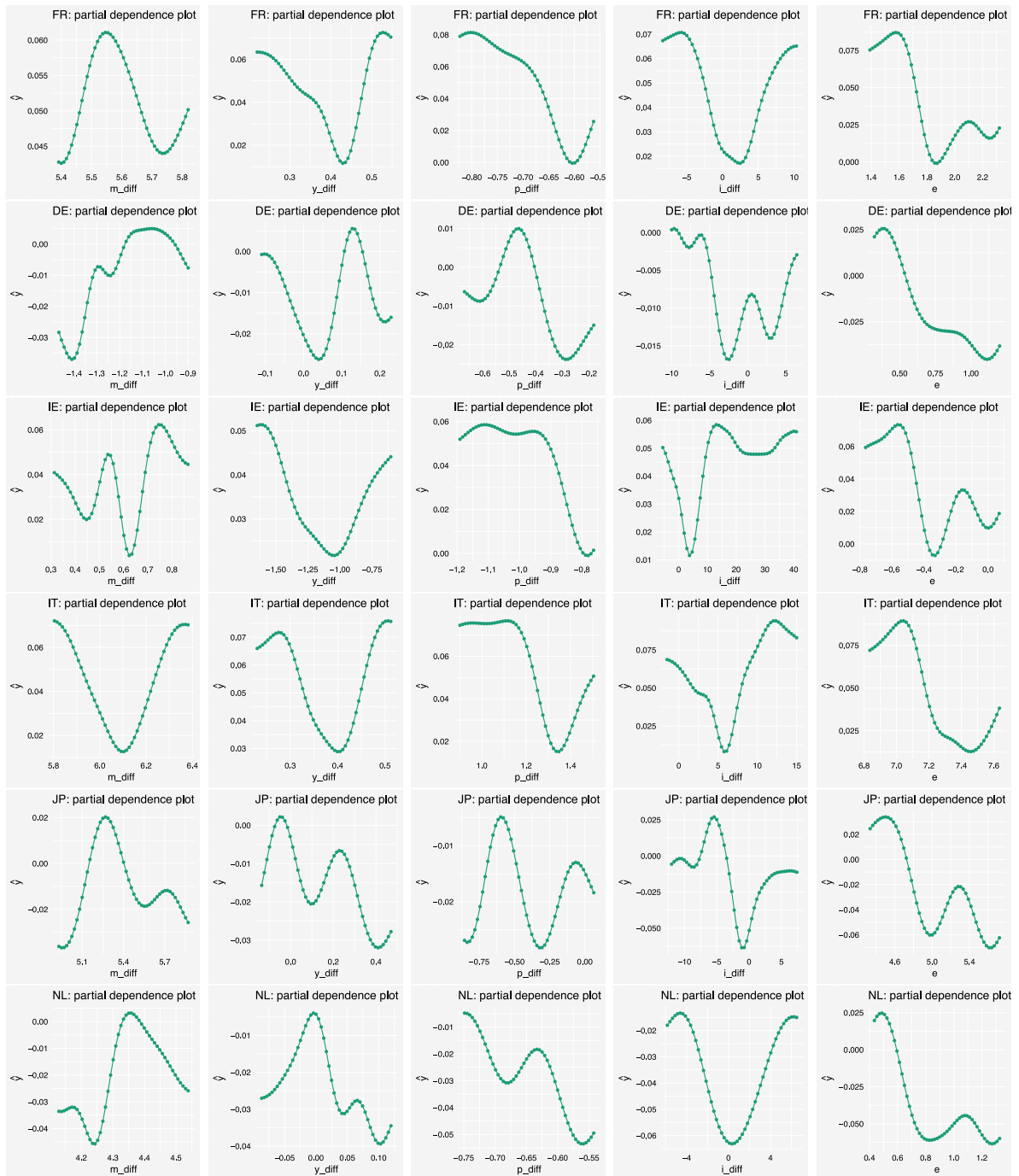


Fig. 25. Partial dependence plots, 12-month forecasts, svm.

a drawbacks of SL tools is that they are somewhat of a black box, and do not provide information economists are accustomed to, such as the standard output from a regression (i.e., coefficients, degree of significance, and goodness of fit).

To analyze the importance of variables in the context of the svm, we started from the more general model used (the monetary model with sticky prices and UIP deviations) in the ECM specification. We performed repeated

permutations on each variable,²⁴ and then calculated loss in terms of the RMSE for the model in which the permuted variable had been used. The higher the increase in the

²⁴ For technical details of the assessment of the importance of the variables by permutation, refer to Breiman (2001, Section 10) Hastie et al. (2009, Section 15.3.2) and Fisher, Rudin, and Dominici (2018).



Fig. 26. Partial dependence plots, 12-month forecasts, svm.

RMSE was when using a permuted variable, the more important the variable was.

Tables 15–18 display the results for the overall sample and the sample divided into decades. Table 15 also contains an additional set of columns indicating, for the sake of clarity, the ranking of the variables in terms of importance. The tables reveal a strong regularity: Across countries, the level of the exchange rate at time t was the most relevant variable, followed by the price differential,

money differential, output differential, and interest rate differential. Interestingly, there were differences across decades; for example, in the 1980s, where differences in monetary policies were stronger, money differentials tended to matter more than in subsequent decades.

The current value of the exchange rate is related to the error correction formulation of the model being estimated. Although the functional form of an svm is completely flexible, and we do not know the specific function

Table 15

Variable importance, overall.

Country	$\hat{m}$	$\hat{y}$	$\hat{p}$	$\hat{i}$	e	$\hat{m}$	$\hat{y}$	$\hat{p}$	$\hat{i}$	e
AU	9.14	9.60	6.18	6.99	13.20	3	2	5	4	1
AT	19.32	17.56	20.44	15.80	27.15	3	4	2	5	1
BE	20.20	14.69	19.17	10.61	21.43	2	4	3	5	1
CA	6.37	7.75	6.26	5.60	7.54	3	1	4	5	2
DK	15.96	16.68	16.31	16.11	17.66	5	2	3	4	1
FI	9.52	8.66	9.88	9.39	12.17	3	5	2	4	1
FR	15.13	12.45	13.90	8.03	17.82	2	4	3	5	1
DE	17.93	13.25	13.80	8.95	15.83	1	4	3	5	2
IE	16.07	13.03	10.36	6.47	19.00	2	3	4	5	1
IT	14.46	13.77	13.62	13.50	15.88	2	3	4	5	1
JP	8.99	8.33	7.86	8.15	9.97	2	3	5	4	1
NL	13.47	16.22	18.55	16.80	21.95	5	4	2	3	1
NZ	21.13	13.58	19.20	14.70	22.60	2	5	3	4	1
PT	14.01	11.93	14.25	10.09	14.89	3	4	2	5	1
ES	8.64	7.71	10.90	4.52	11.50	3	4	2	5	1
SE	6.12	6.61	8.97	8.26	10.67	5	4	2	3	1
CH	10.22	6.64	12.56	12.58	11.51	4	5	2	1	3
GB	4.96	4.61	5.44	5.33	6.69	4	5	2	3	1
Mean	12.87	11.28	12.65	10.10	15.41	3	3.7	2.9	4.2	1.2

Table 16

Variable importance, '80s.

Country	$\hat{m}$	$\hat{y}$	$\hat{p}$	$\hat{i}$	e	$\hat{m}$	$\hat{y}$	$\hat{p}$	$\hat{i}$	e
AU	6.20	5.97	7.08	4.94	7.35	3	4	2	5	1
AT	30.16	23.14	24.22	23.81	30.77	2	5	3	4	1
BE	31.11	24.48	30.13	13.97	28.95	1	4	2	5	3
CA	12.32	9.13	14.88	8.80	15.89	3	4	2	5	1
FI	17.82	10.64	19.63	13.51	18.40	3	5	1	4	2
FR	16.31	19.31	17.72	10.39	19.91	4	2	3	5	1
DE	26.85	18.60	18.99	11.18	19.75	1	4	3	5	2
IE	25.55	20.24	20.32	10.79	26.86	2	4	3	5	1
IT	27.98	24.92	24.74	21.03	28.37	2	3	4	5	1
JP	19.81	19.86	18.71	14.95	19.13	2	1	4	5	3
NL	25.08	20.84	22.61	24.09	26.41	2	5	4	3	1
NZ	15.14	14.24	13.23	13.37	17.28	2	3	5	4	1
PT	26.92	23.30	29.18	18.60	27.82	3	4	1	5	2
ES	13.19	10.59	20.36	5.83	14.39	3	4	1	5	2
SE	16.94	11.69	15.69	10.82	16.05	1	4	3	5	2
GB	19.44	14.81	18.68	16.71	18.98	1	5	3	4	2
Mean	20.68	16.98	19.76	13.92	21.02	2.19	3.81	2.75	4.63	1.63

Table 17

Variable importance, '90s.

Country	$\hat{m}$	$\hat{y}$	$\hat{p}$	$\hat{i}$	e	$\hat{m}$	$\hat{y}$	$\hat{p}$	$\hat{i}$	e
AU	19.63	18.79	18.00	16.25	18.16	1	2	4	5	3
AT	12.59	15.67	15.77	14.26	16.79	5	3	2	4	1
BE	11.34	11.11	10.60	10.62	11.74	2	3	5	4	1
CA	7.70	7.04	8.03	6.78	9.16	3	4	2	5	1
FI	11.51	8.25	8.42	11.65	11.83	3	5	4	2	1
FR	7.38	7.62	7.04	6.85	7.87	3	2	4	5	1
DE	13.61	15.90	16.44	14.10	15.74	5	2	1	4	3
IE	6.22	5.62	5.91	3.12	6.70	2	4	3	5	1
IT	6.08	8.85	8.44	8.71	9.20	5	2	4	3	1
JP	11.21	11.87	11.39	10.60	12.92	4	2	3	5	1
NL	13.73	14.72	12.74	15.53	14.92	4	3	5	1	2
NZ	33.32	21.11	22.90	20.10	24.56	1	4	3	5	2
PT	6.32	7.83	7.25	6.93	8.27	5	2	3	4	1
ES	12.80	12.18	14.35	13.38	15.21	4	5	2	3	1
SE	7.75	5.59	6.11	4.23	8.05	2	4	3	5	1
GB	6.35	6.64	7.74	8.52	10.37	5	4	3	2	1
Mean	11.72	11.17	11.32	10.73	12.59	3.38	3.19	3.19	3.88	1.38

Table 18
Variable importance: 2000s.

Country	$\hat{m}$	$\hat{y}$	$\hat{p}$	$\hat{i}$	e	$\hat{m}$	$\hat{y}$	$\hat{p}$	$\hat{i}$	e
AU	15.78	14.17	14.99	13.19	18.46	2	4	3	5	1
CA	10.40	11.23	9.65	8.26	11.24	3	2	4	5	1
DK	15.96	16.68	16.31	16.11	17.66	5	2	3	4	1
JP	10.07	8.08	9.15	6.86	9.02	1	4	2	5	3
NZ	21.66	15.20	18.74	19.26	22.86	2	5	4	3	1
SE	13.87	14.54	15.31	18.23	18.79	5	4	3	2	1
CH	8.13	5.06	8.41	8.23	8.30	4	5	1	3	2
GB	5.36	4.76	5.99	4.59	5.73	3	4	1	5	2
Mean	12.65	11.21	12.32	11.84	14.01	3.13	3.75	2.63	4.00	1.50

that the model is estimating, the aim of the ECM specification is to capture the relevance of a long-run relationship between the relevant variables. In this sense, we can interpret the relevance of the current level of the exchange rate as follows: the long-run equilibrium value of the exchange rate constrains the forecasted values (considering that the forecasts in this case were 12 months ahead). This is also supported by the relevance of the price differentials, which is in line with the importance of deviations from the PPP, another long-run equilibrium relationship.

The evidence so far refers to the importance of variables in a sample (permutations were implemented in the estimation phase). Fig. 23 reports the information for the out-of-sample importance of the variables, where permutations were implemented on the value of the variables used in forecasting (i.e., on each element of the vector X_t). The results broadly confirm the findings presented in Table 15. In particular, the exchange rate differential was still the most relevant variable, with an average ranking of 1.8 followed by money differential and price differential with average rankings of 2.8 and 3, respectively.

5.6. Variable interactions

A second way of investigating the functioning of SL methods is by using partial dependence plots. These plots display the marginal relation between each variable and the outcome variable (i.e., how the relationship between each variable x and the outcome changes at different values of x). In general, a certain amount of caution needs to be exercised in interpreting partially dependent plots. The variables of the models that we implemented were all interdependent and, ideally, a multi-dimensional graph should be used to capture the complex linkages between them. However, this comes at the cost of losing the ability to explain the functioning of statistical learning tools. However, based on simplifying assumptions, partially dependent plots can be instructive of the type and form of the relationship between the fundamentals and the exchange rate, as implied by SL models, and are a first step toward a better explanation of those tools.

Figs. 24–26 display the results, suggesting some interesting insights. First, the non-linearity that SL models captured emerges clearly. Second, within a highly non-linear pattern, it is possible to identify some recurring patterns. In several countries, there emerged a negative relationship between Δe_t and e_t , suggesting a form of adjustment in the behavior of the exchange rate. Moreover,

the U-shaped relationship between the interest differential and Δe_t was in line with UIP²⁵ and, most importantly, with the empirical evidence supporting non-linear adjustment from deviations in the UIP (Sarno et al., 2006). The relationship between price differentials and future exchange rate change was generally negative, whereas no clear pattern emerged for money and output differentials.

6. Conclusions

This study used innovative tools recently developed in the statistical learning literature to assess the capability of standard exchange rate models in predicting the exchange rate in the short and long runs. Our results show that SL tools have remarkably good predicting capabilities, outperforming the random walk in forecasting the exchange rate at different forecasting horizons, with the exception of the very short term (one to two months). In addition to forecasting performance, our study shows that SL can be used to improve our understanding of the relationship between the exchange rate and economic fundamentals. Fundamentals, as contained in standard exchange rate models, do matter for forecasting the exchange rate, but the functional form and method of estimation used in the standard literature are too restrictive, and do not permit the capturing of the complexity of the relationship between the exchange rate and fundamentals. Our paper sheds light on the relationship between the variables implied by the statistical learning models, although much work has to be done to improve the understanding of how SL tools work. This is a major issue in the field of Artificial Intelligence²⁶; its solution will certainly reverberate throughout the social sciences as well.

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²⁵ A positive interest differential is positively correlated with an appreciation of the dollar and a negative differential with a depreciation.

²⁶ See the DARPA initiative at <https://www.darpa.mil/program/explainable-artificial-intelligence>.

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